



Multimodal Machine Learning

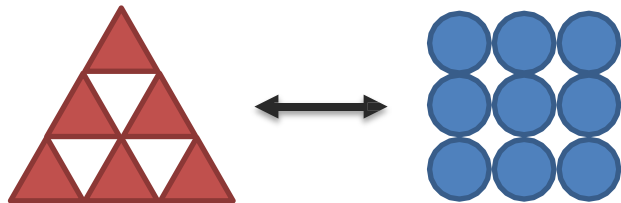
Lecture 12.2: Quantification

Dam Quang Tuan

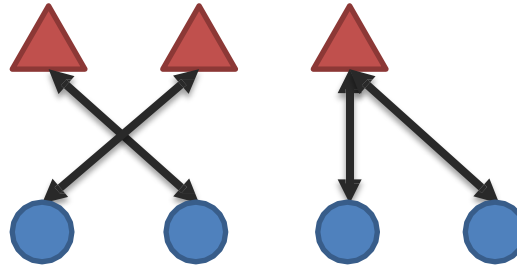
Quantification

Definition: Empirical and theoretical study to better understand heterogeneity, cross-modal interactions, and the multimodal learning process.

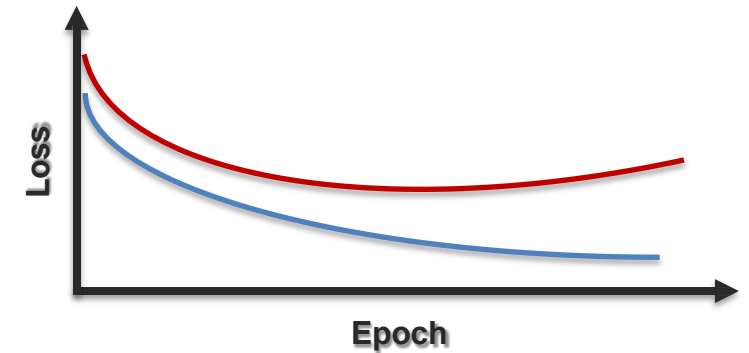
(A) Heterogeneity



(B) Interactions

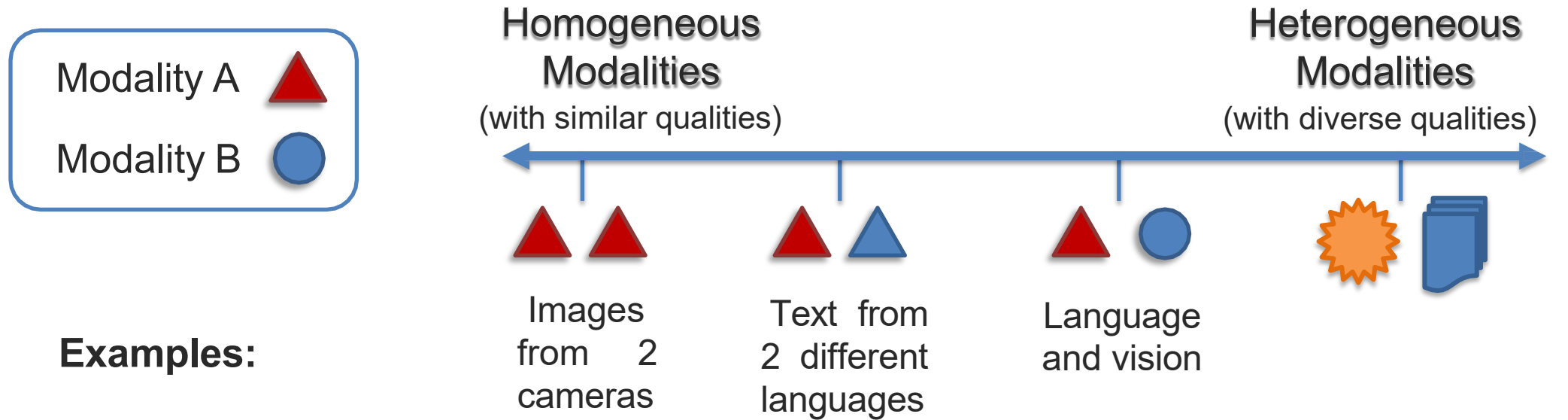


(C) Learning



Sub-Challenge 6a: Heterogeneity

Definition: Quantifying the dimensions of heterogeneity in multimodal datasets and how they subsequently influence modeling and learning.



① Element representation

② Element distribution

③ Structure

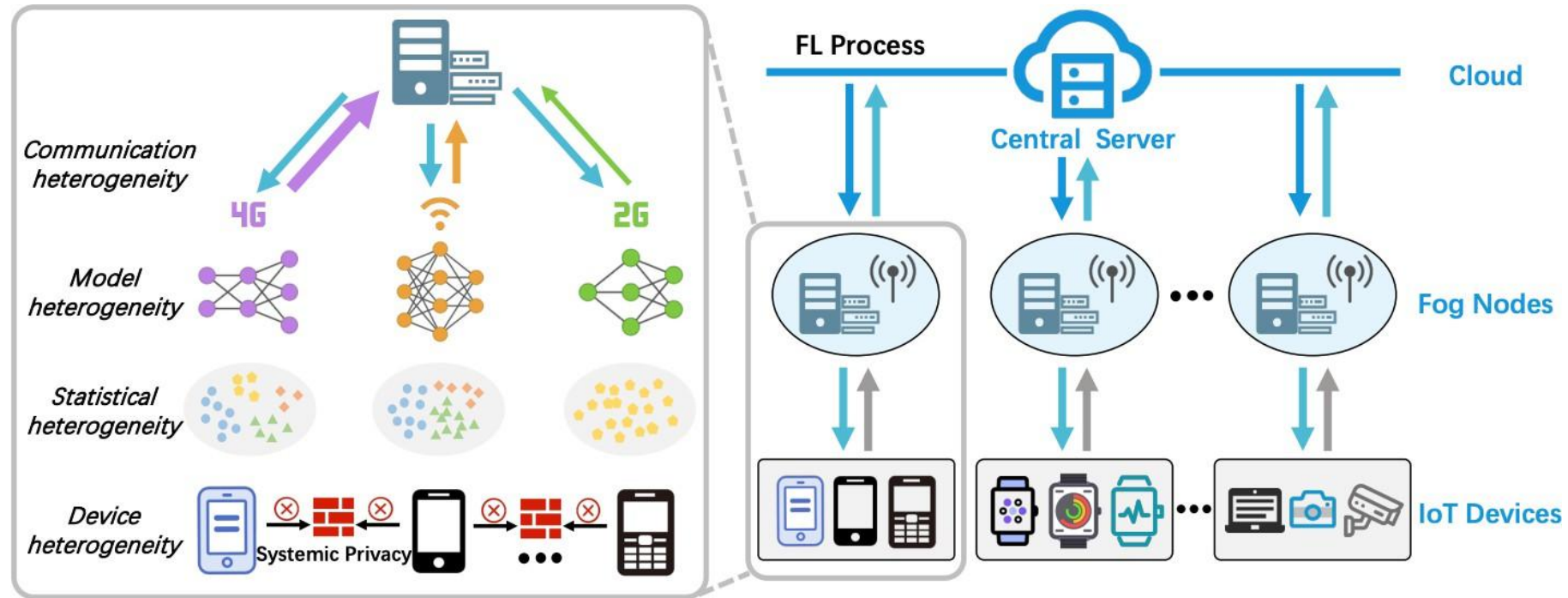
④ Information

⑤ Noise

⑥ Relevance

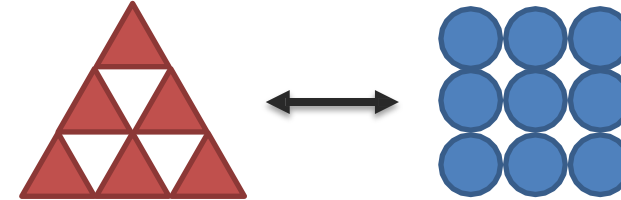
Distribution heterogeneity

Inspired by distributed learning



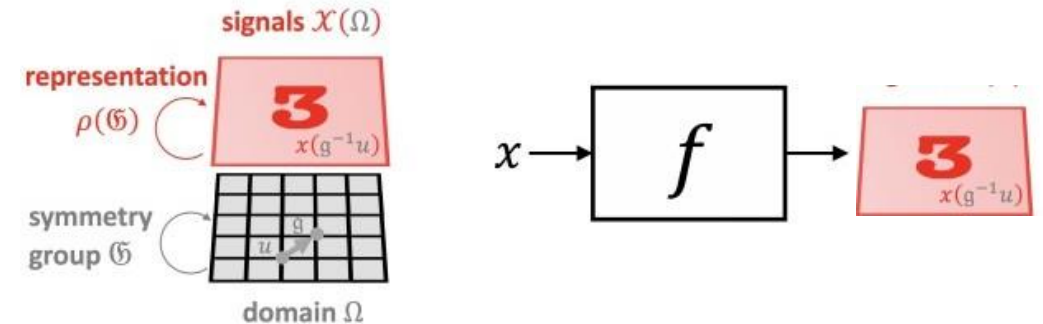
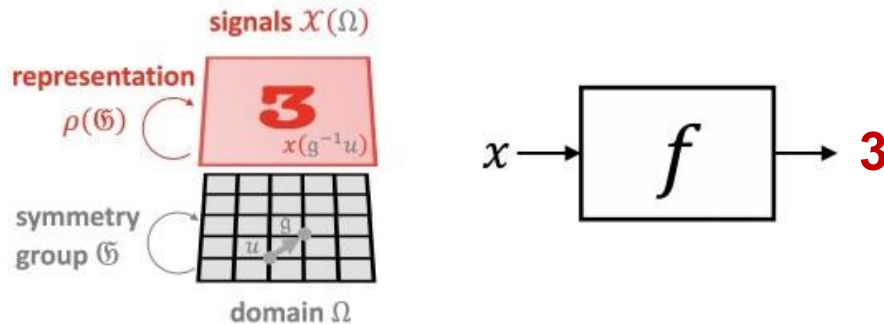
Structure heterogeneity

Inspired by structure learning



A function $f : \mathcal{X}(\Omega) \rightarrow \mathcal{Y}$ is $\mathfrak{G}$ -invariant if $f(\rho(\mathfrak{g})x) = f(x)$ for all $\mathfrak{g} \in \mathfrak{G}$ and $x \in \mathcal{X}(\Omega)$, i.e., its output is unaffected by the group action on the input.

A function $f : \mathcal{X}(\Omega) \rightarrow \mathcal{X}(\Omega)$ is $\mathfrak{G}$ -equivariant if $f(\rho(\mathfrak{g})x) = \rho(\mathfrak{g})f(x)$ for all $\mathfrak{g} \in \mathfrak{G}$, i.e., group action on the input affects the output in the same way.



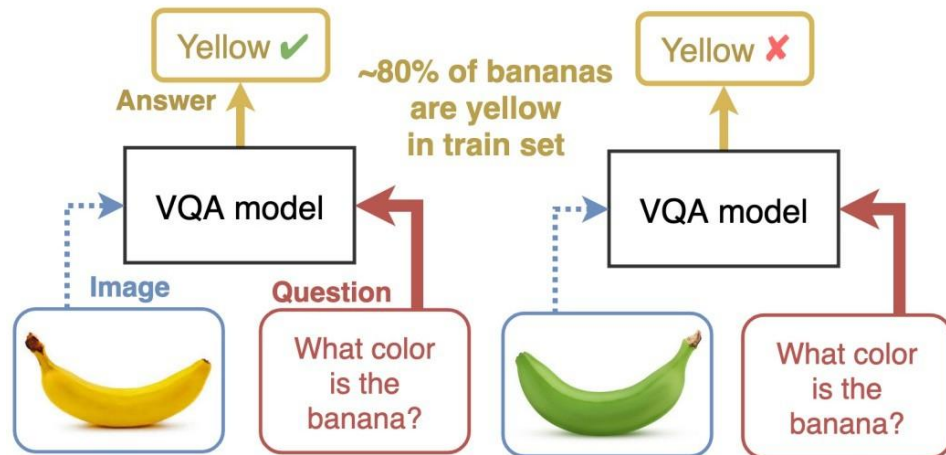
Derives deep learning architectures on grids, graphs, sets, etc.

Modality Biases

Heterogeneity in information and relevance

Unimodal biases and modality collapse

VQA models answer the question without looking at the image

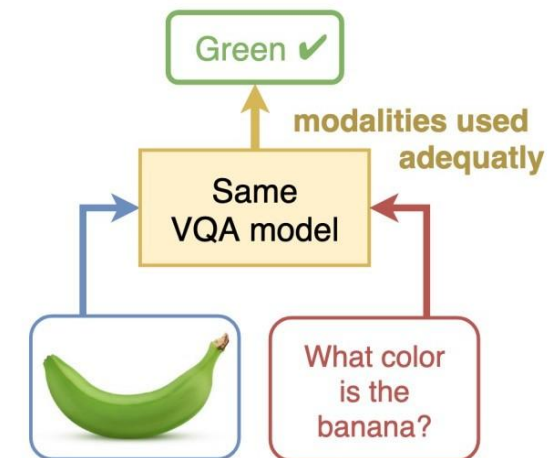


Balancing modalities

Balancing training



Not the case when trained with RUBi



[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

[Javaloy et al., Mitigating Modality Collapse in Multimodal VAEs via Impartial Optimization. ICML 2022]

[Goyal et al., Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering. CVPR 2017]

Modality Biases

Heterogeneity in information and relevance

Fairness and social biases – unimodal social biases

Finding: Image captioning models capture spurious correlations between gender and generated actions

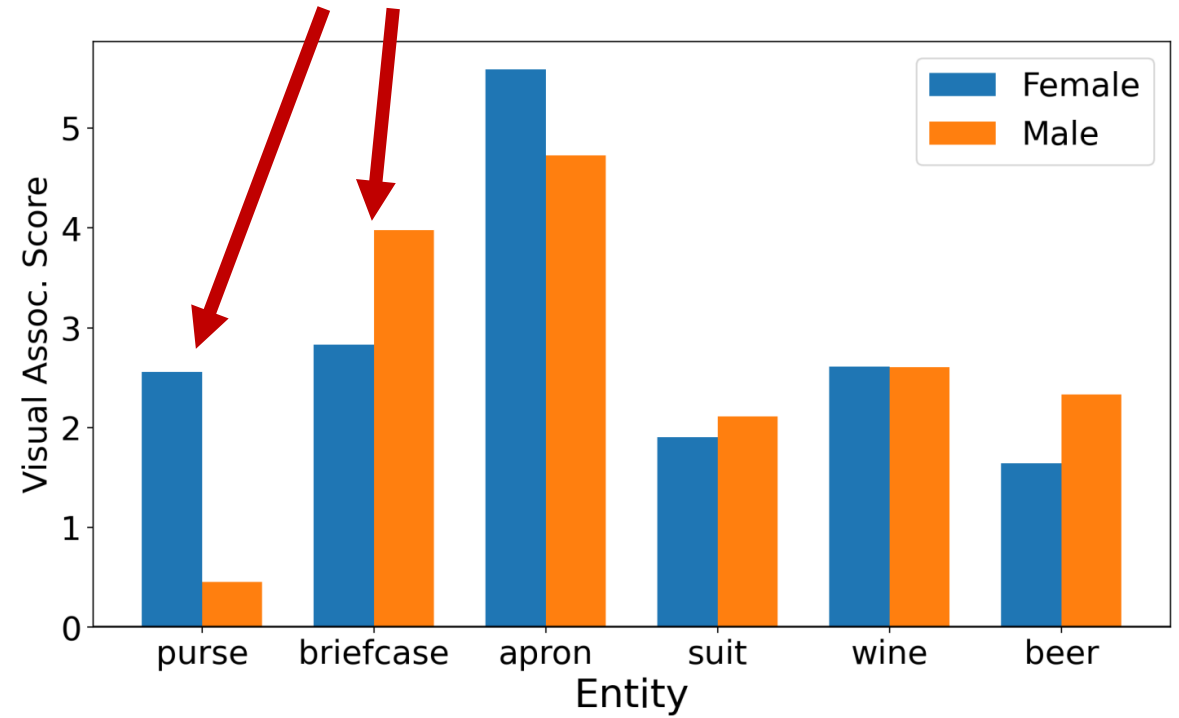
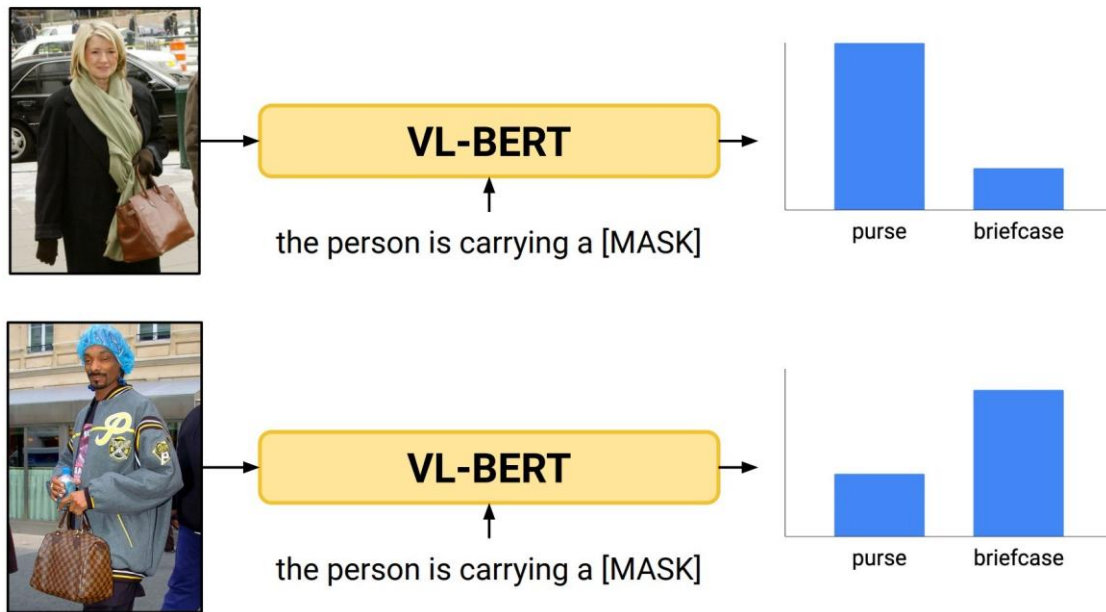


Modality Biases

Heterogeneity in information and relevance

Fairness and social biases – cross-modal interactions worsen social biases

Visual information makes model more confident
in reinforcing gender stereotypes

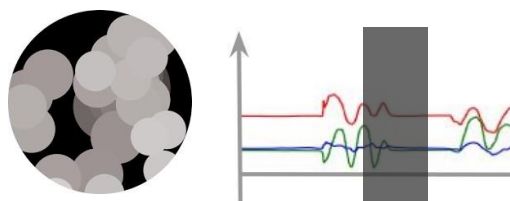


Noise Topologies and Robustness

Heterogeneity in noise

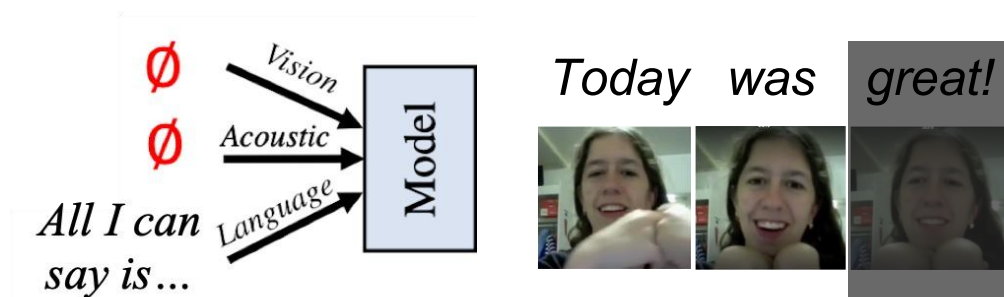
Modality-specific robustness

noise → **nosie**



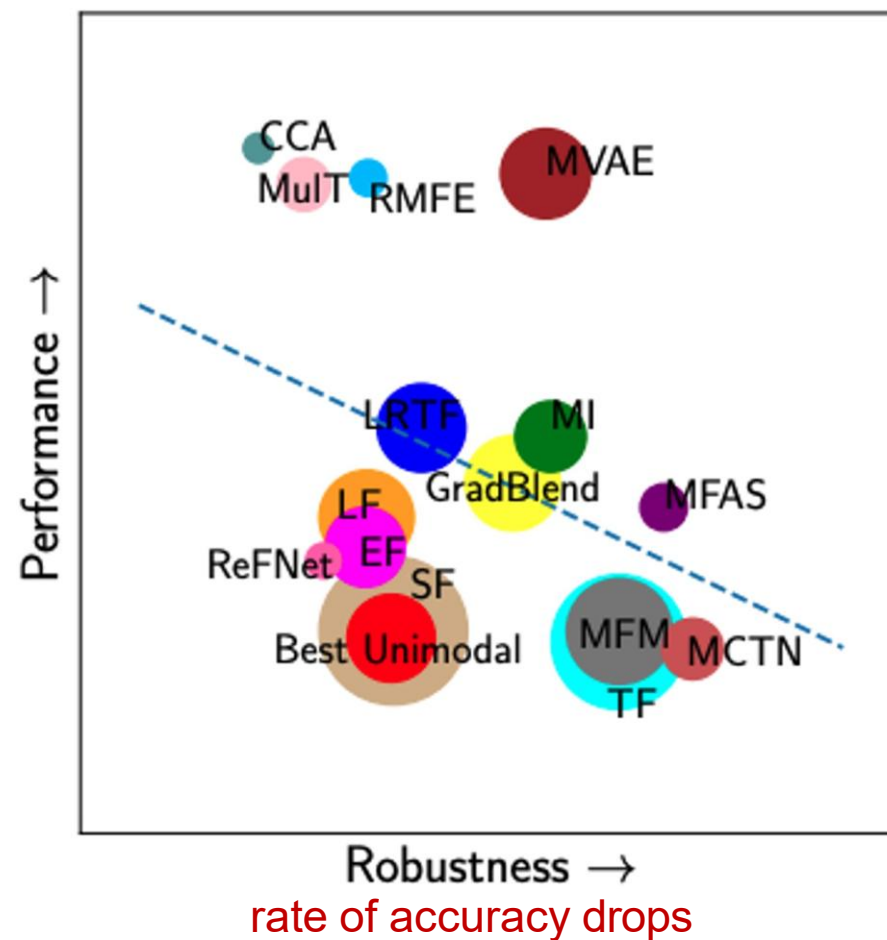
[Belinkov & Bisk, 2018; Subramaniam et al., 2009; Boyat & Joshi, 2015]

Multimodal robustness



[Zadeh et al., 2020]

Strong tradeoffs between performance and robustness

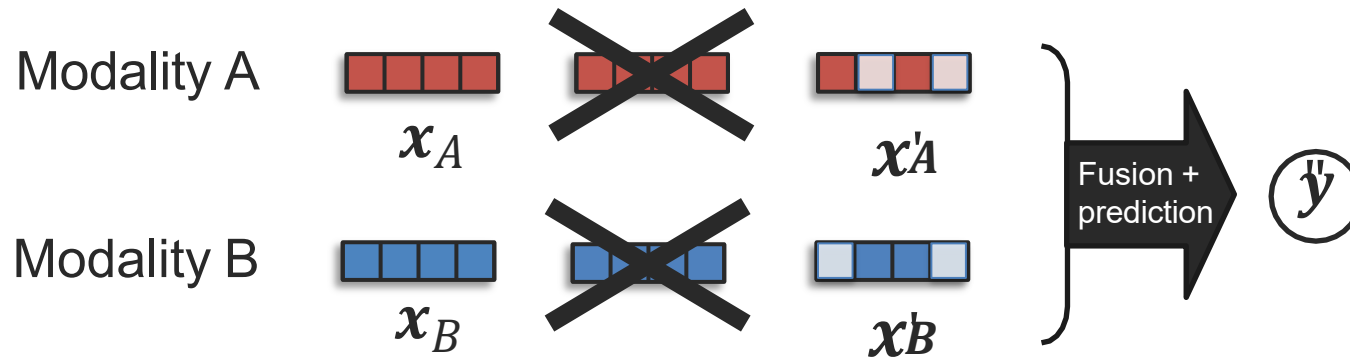


[Liang et al., MultiBench: Multiscale Benchmarks for Multimodal Representation Learning. NeurIPS 2021]

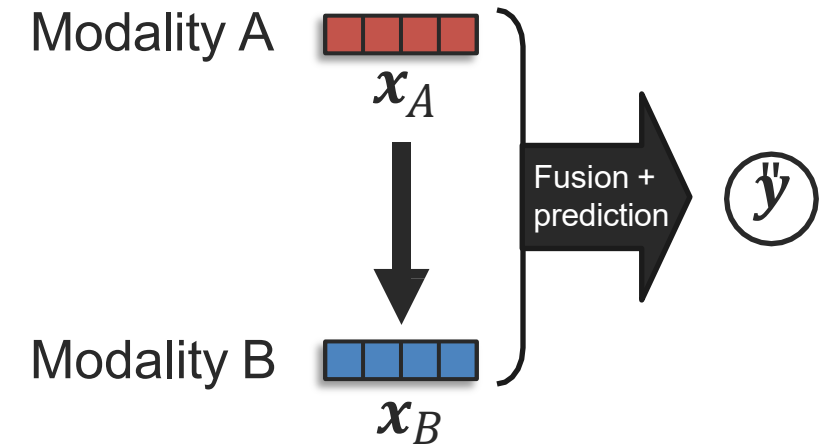
Noise Topologies and Robustness

Several approaches towards more robust models

Robust data + training



Infer missing modalities



Translation model
Joint probabilistic model

[Ngiam et al., Multimodal Deep Learning. ICML 2011]

[Srivastava and Salakhutdinov, Multimodal Learning with Deep Boltzmann Machines. JMLR 2014]

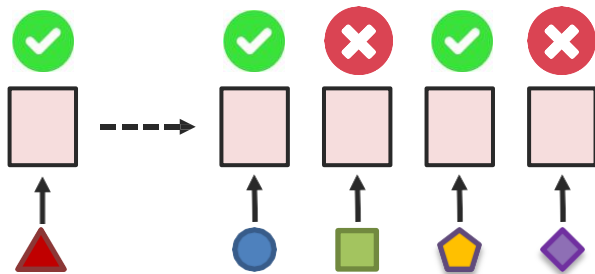
[Tran et al., Missing Modalities Imputation via Cascaded Residual Autoencoder. CVPR 2017]

[Pham et al., Found in Translation: Learning Robust Joint Representations via Cyclic Translations Between Modalities. AAAI 2019]

Quantifying Heterogeneity via Transfer

Information transfer, transfer learning perspective

1a. Estimate modality heterogeneity via transfer



In practice, efficient by pre-trained models and few-shot transfer

Implicitly captures these:

① Element representation

② Element distribution

③ Structure

④ Information

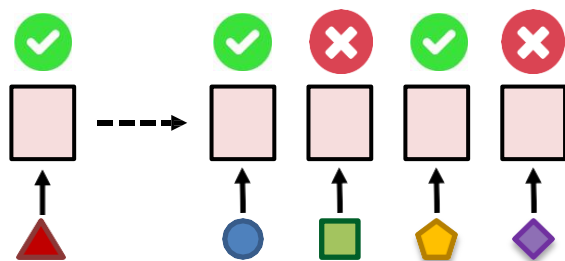
⑤ Noise

⑥ Relevance

Heterogeneity-aware Fusion

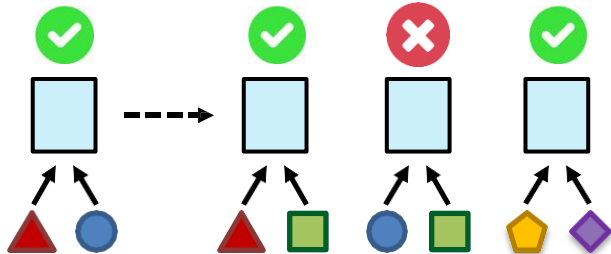
Information transfer, transfer learning perspective

1a. Estimate modality heterogeneity via transfer



(Implicitly captures heterogeneity)

1b. Estimate interaction heterogeneity via transfer



2a. Compute modality heterogeneity matrix

	▲	●	■	⬠	◆
▲	0				
●	1	0			
■	3	2	0		
⬠	1	2	3	0	
◆	5	4	6	3	0

2b. Compute interaction heterogeneity matrix

	{▲●}	{▲■}	{●■}	{⬠◆}
{▲●}	0			
{▲■}	1	0		
{●■}	3	2	0	
{⬠◆}	1	2	4	0

3. Determine parameter clustering

$$\mathbb{U}_1 = \{U_1, U_2, U_4\}$$

$$\mathbb{U}_2 = \{U_3\}$$

$$\mathbb{U}_3 = \{U_5\}$$

$$\mathbb{C}_1 = \{C_{12}, C_{13}, C_{45}\}$$

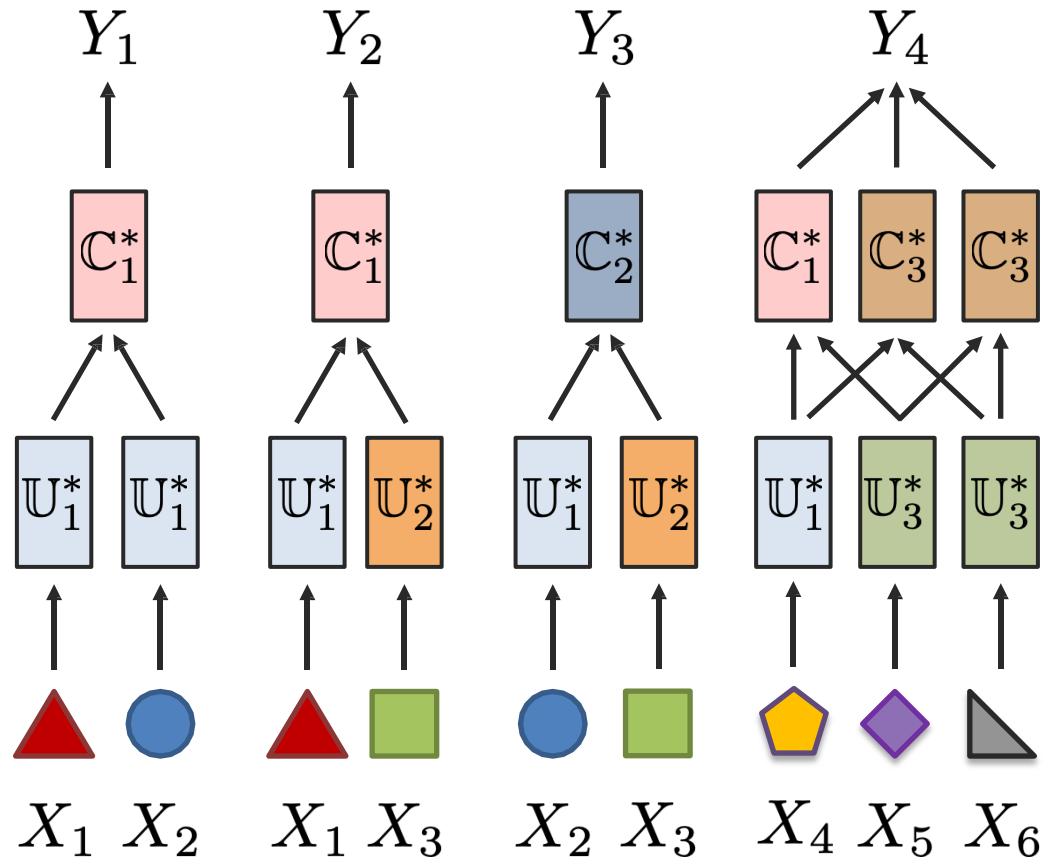
$$\mathbb{C}_2 = \{C_{23}\}$$

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

[Liang et al., HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

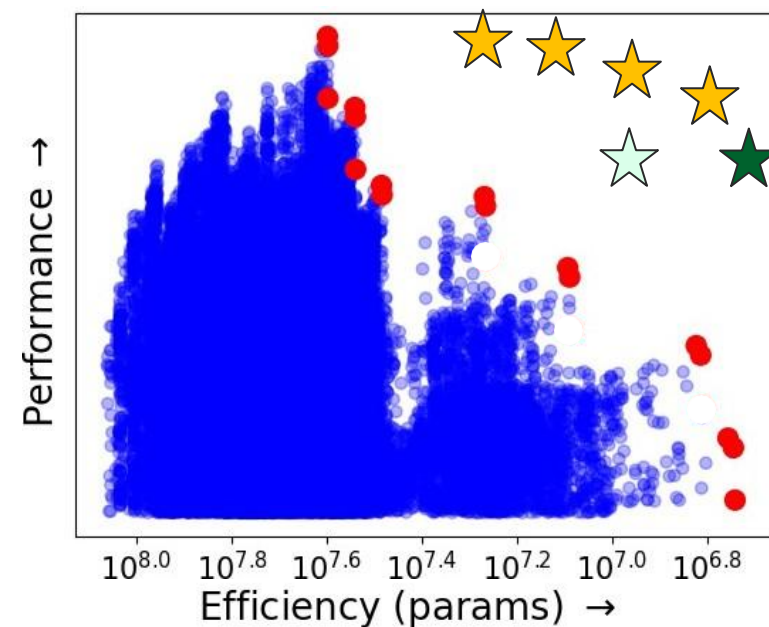
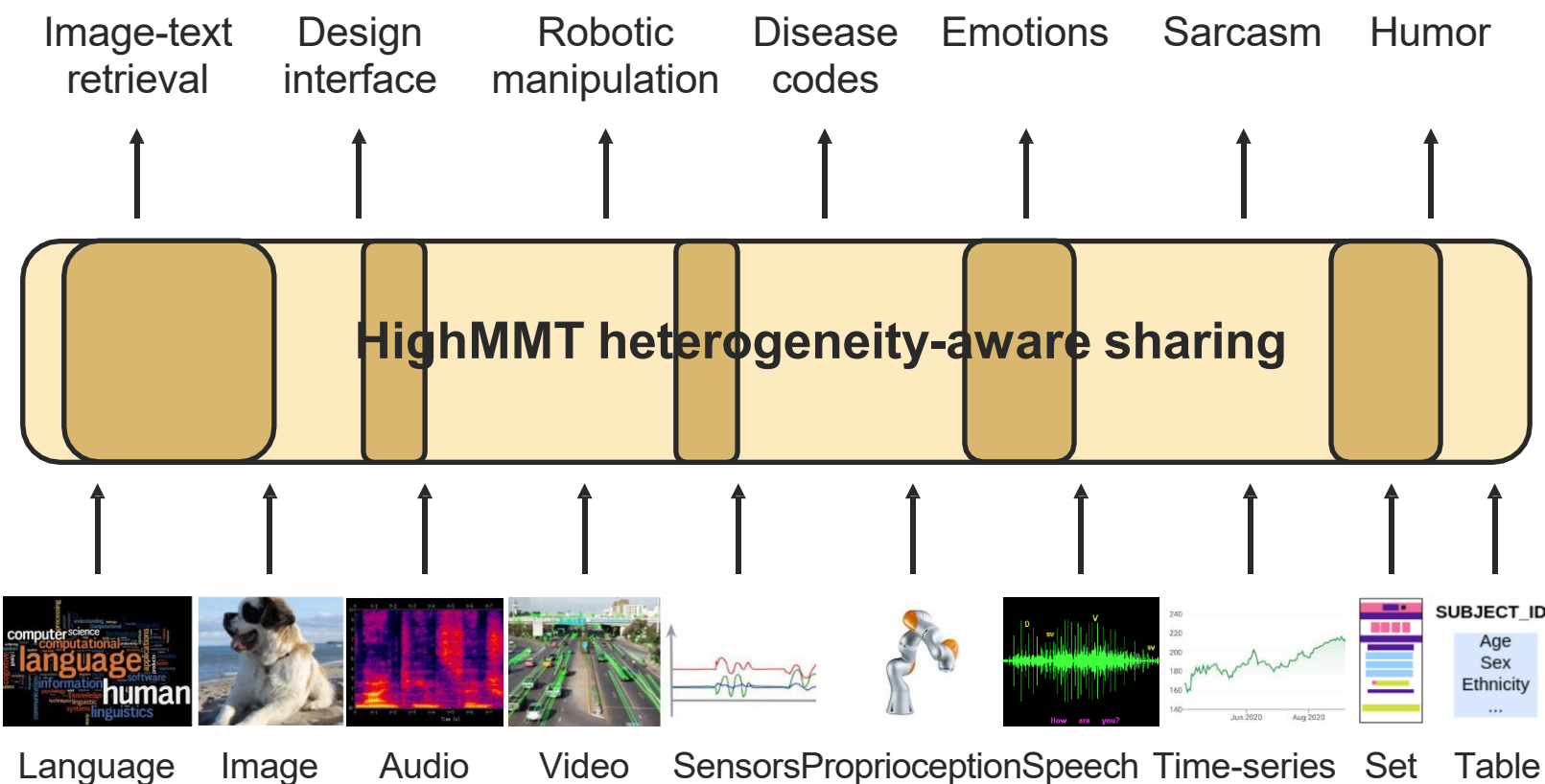
Heterogeneity-aware Fusion

Information transfer, transfer learning perspective



Quantifying Modality Heterogeneity

HighMMT heterogeneity-aware: estimate heterogeneity to determine parameter sharing



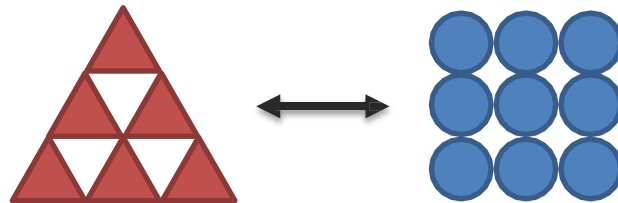
- All model combinations (>10,000)
- Pareto front
- HighMMT single-task
- HighMMT multitask
- **HighMMT heterogeneity-aware**

Challenges: Quantifying Heterogeneity

Open
challenges

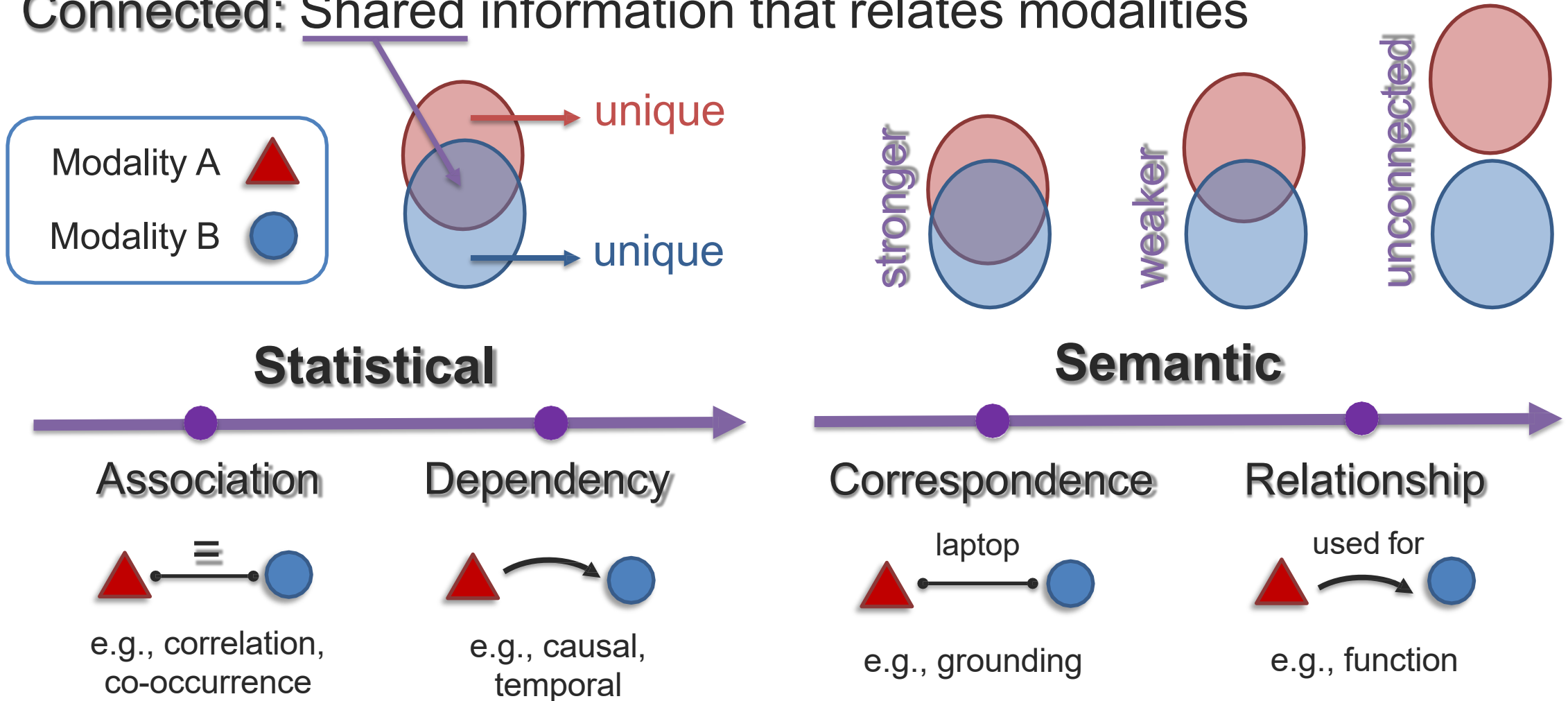
Open challenges:

- Noisy and missing modalities.
- New and understudied modalities.
- Large number of modalities.
- Cases where its unclear which modalities are useful – active selection
- Related fields: federated learning, active learning, distributed systems, structure & invariances



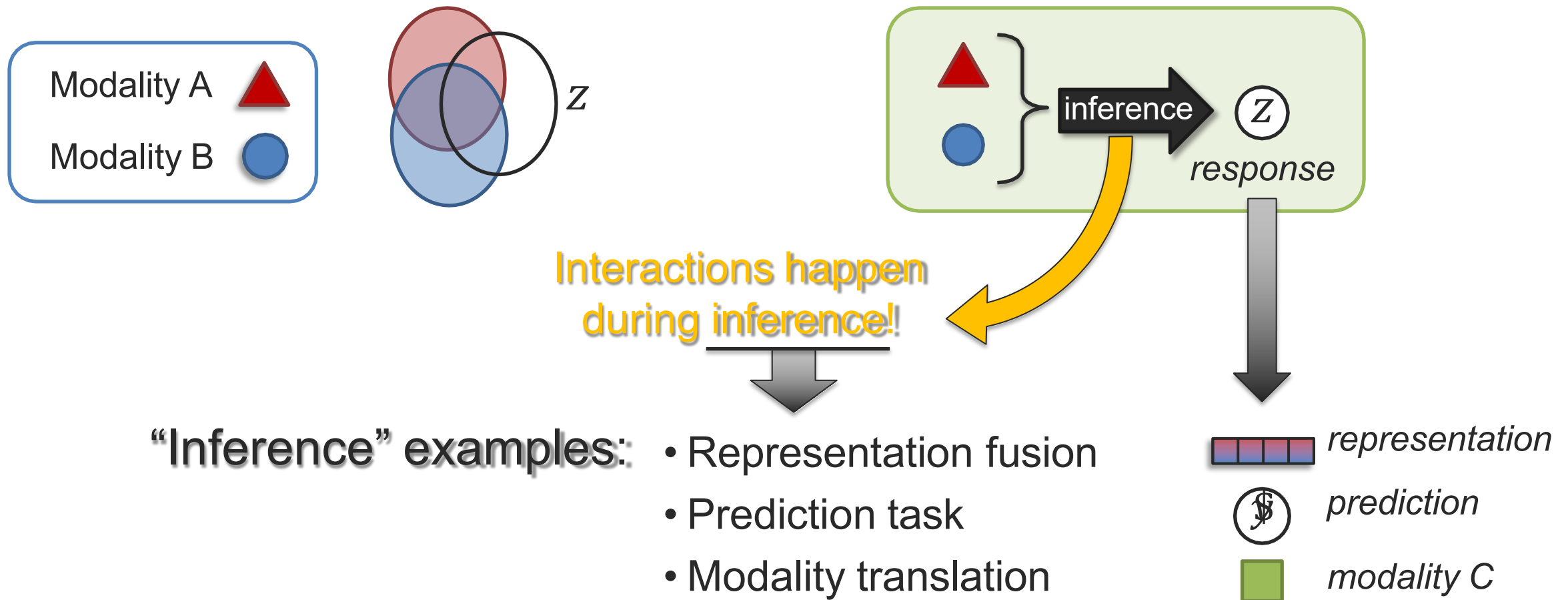
Sub-Challenge 6b: Cross-modal Connections

Connected: Shared information that relates modalities



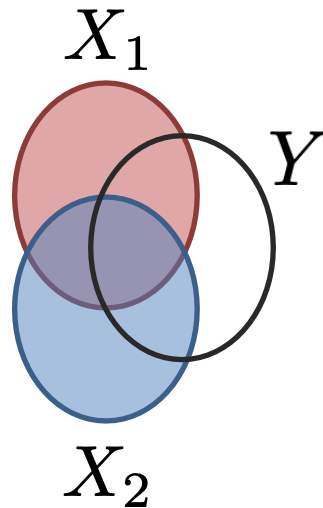
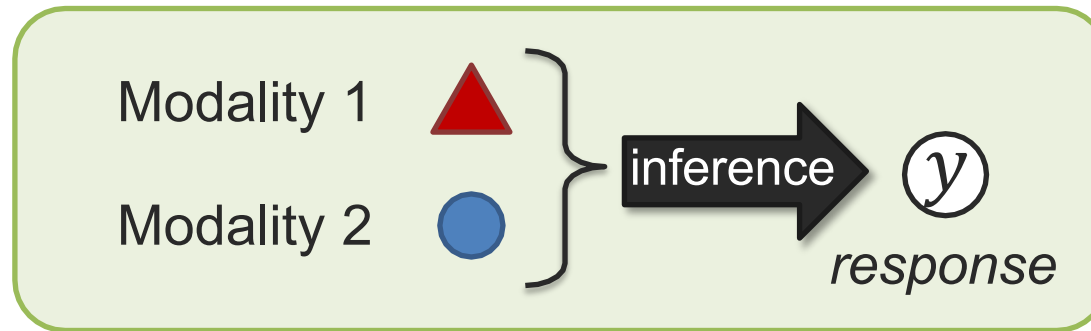
Sub-Challenge 6b: Cross-modal Interactions

Interacting: process affecting each modality, creating new response



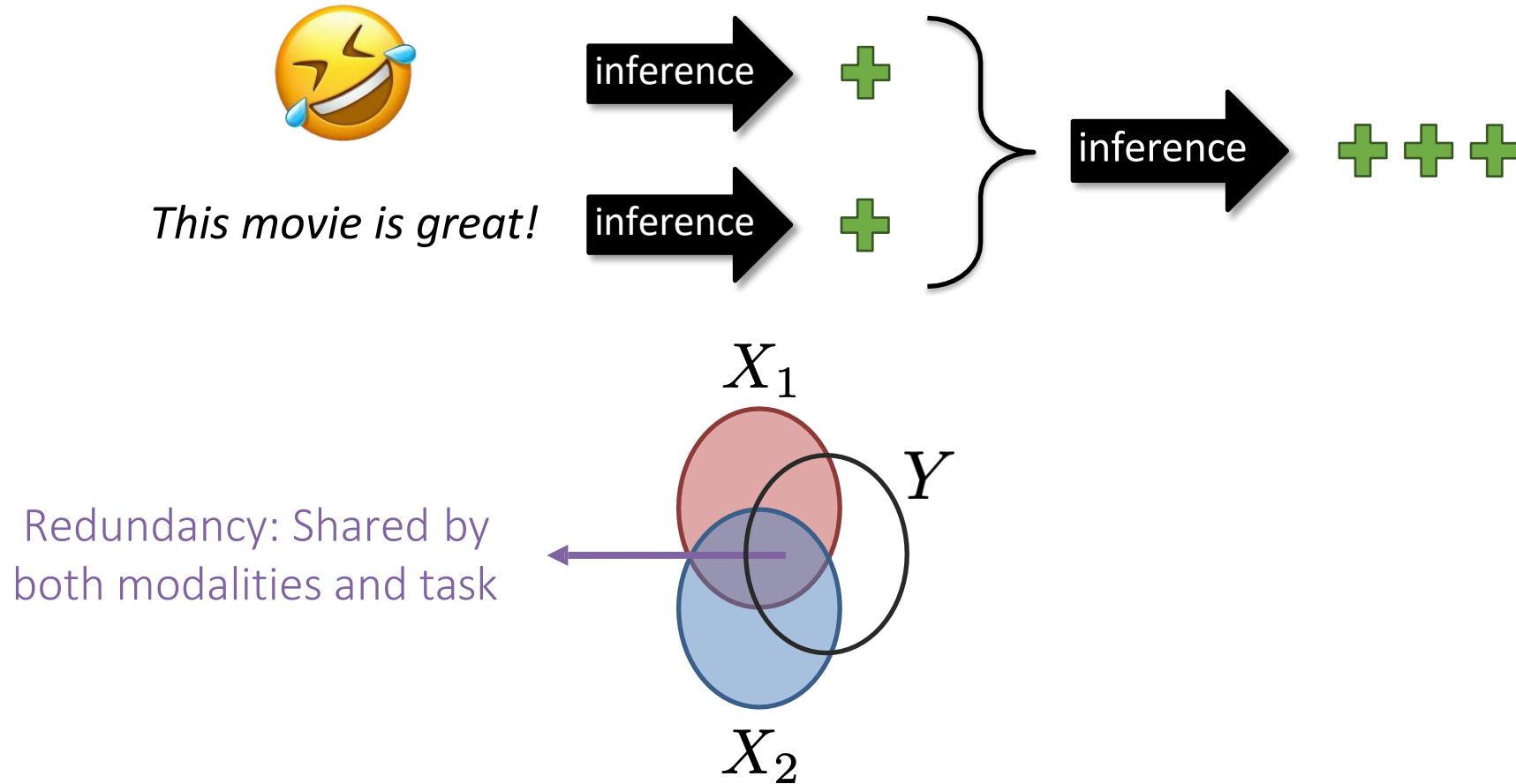
Part 1: Multimodal Interactions

Interactions: Understanding *commonalities* between modalities and how they *combine* to provide information for a task.



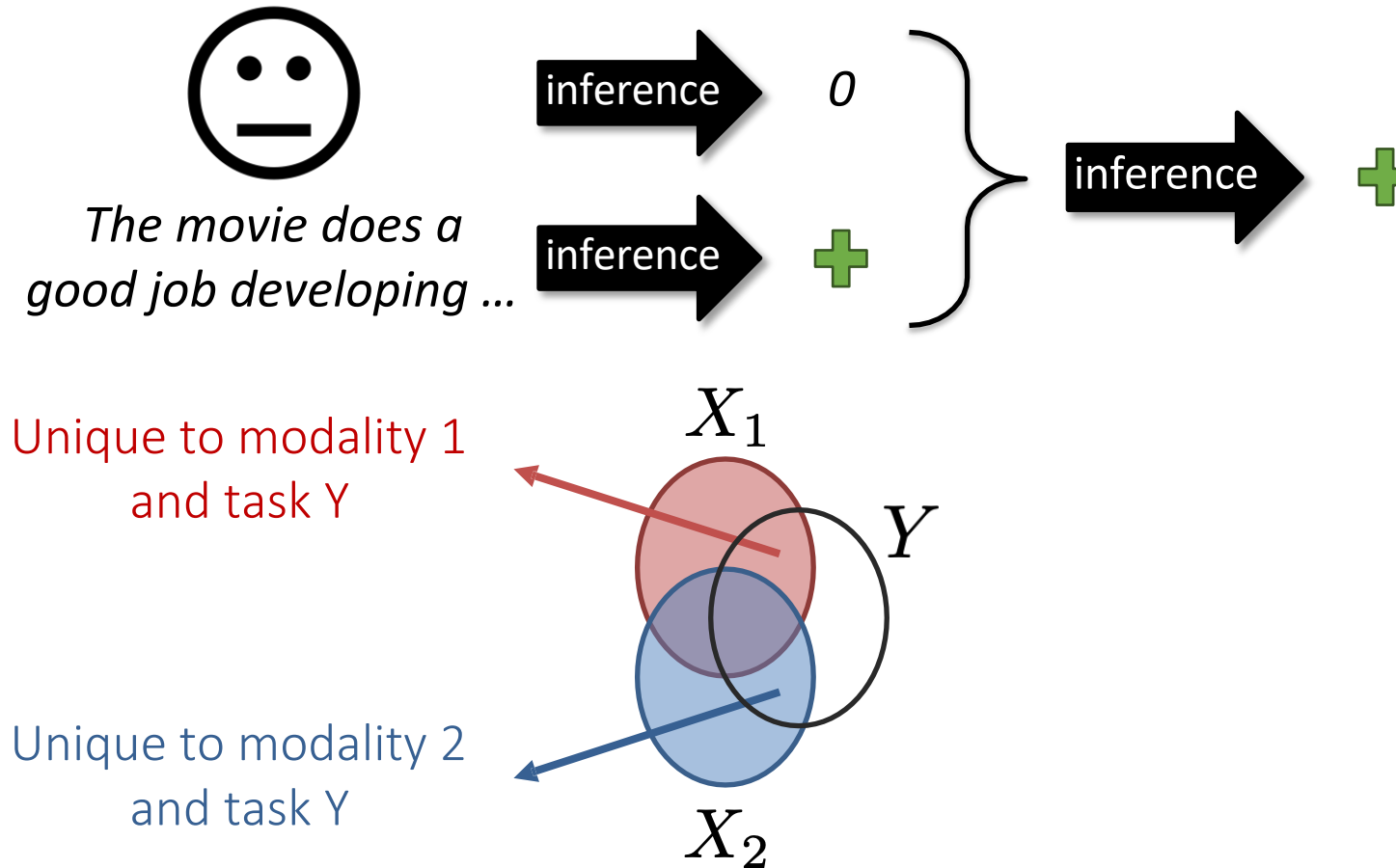
Multimodal Interactions

Interactions: Understanding *commonalities* between modalities and how they *combine* to provide information for a task.



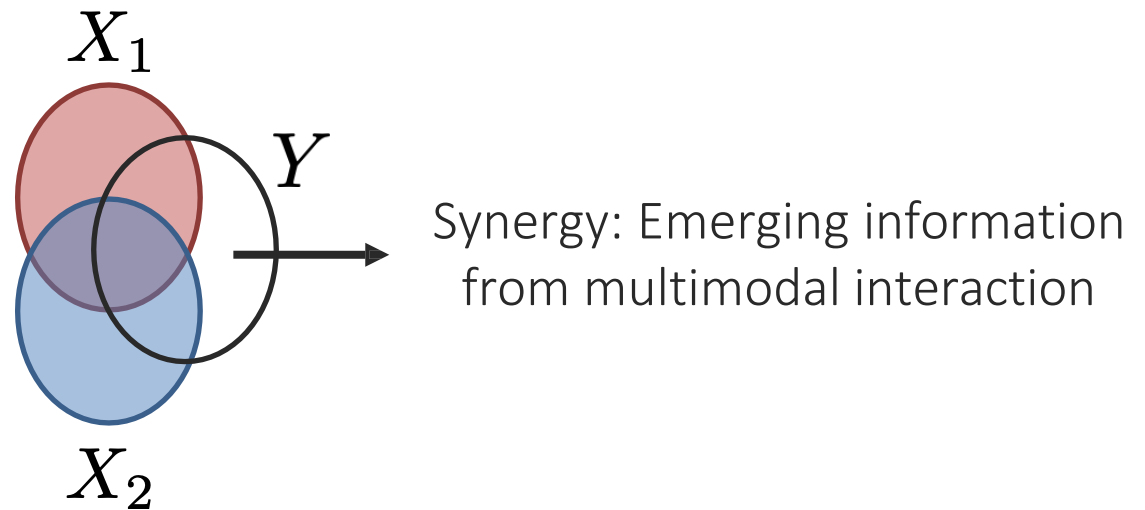
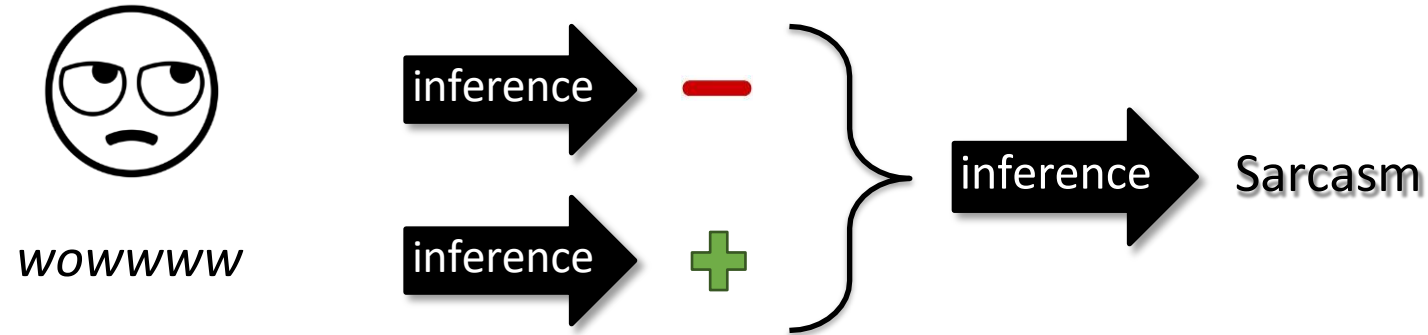
Multimodal Interactions

Interactions: Understanding *commonalities* between modalities and how they *combine* to provide information for a task.



Multimodal Interactions

Interactions: Understanding *commonalities* between modalities and how they *combine* to provide information for a task.



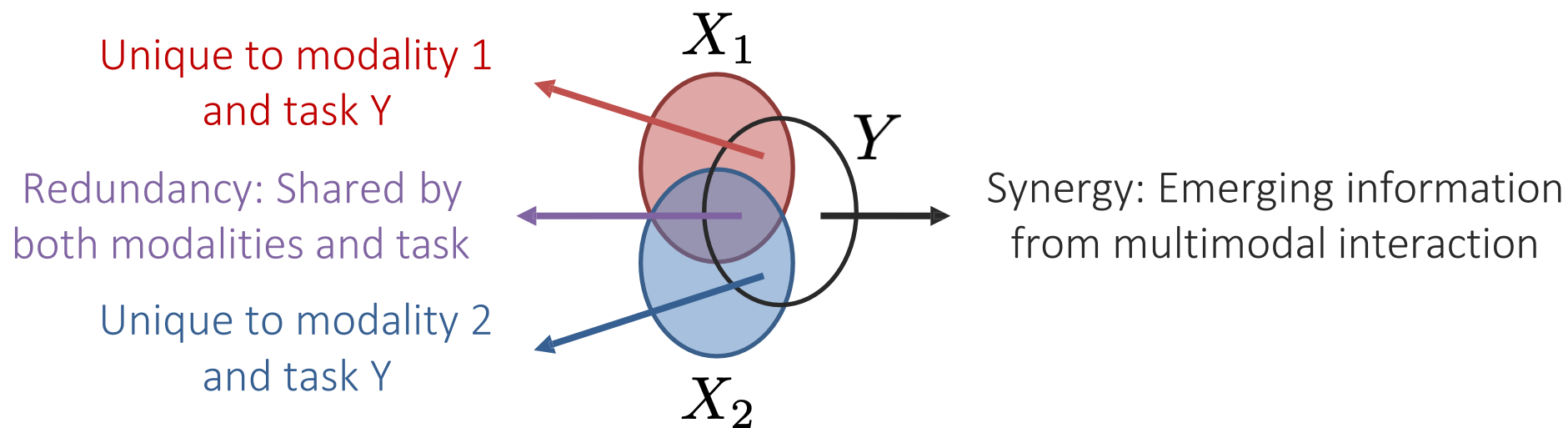
Quantifying Multimodal Interactions

Fundamental questions in multimodal learning

What interactions are in my data?

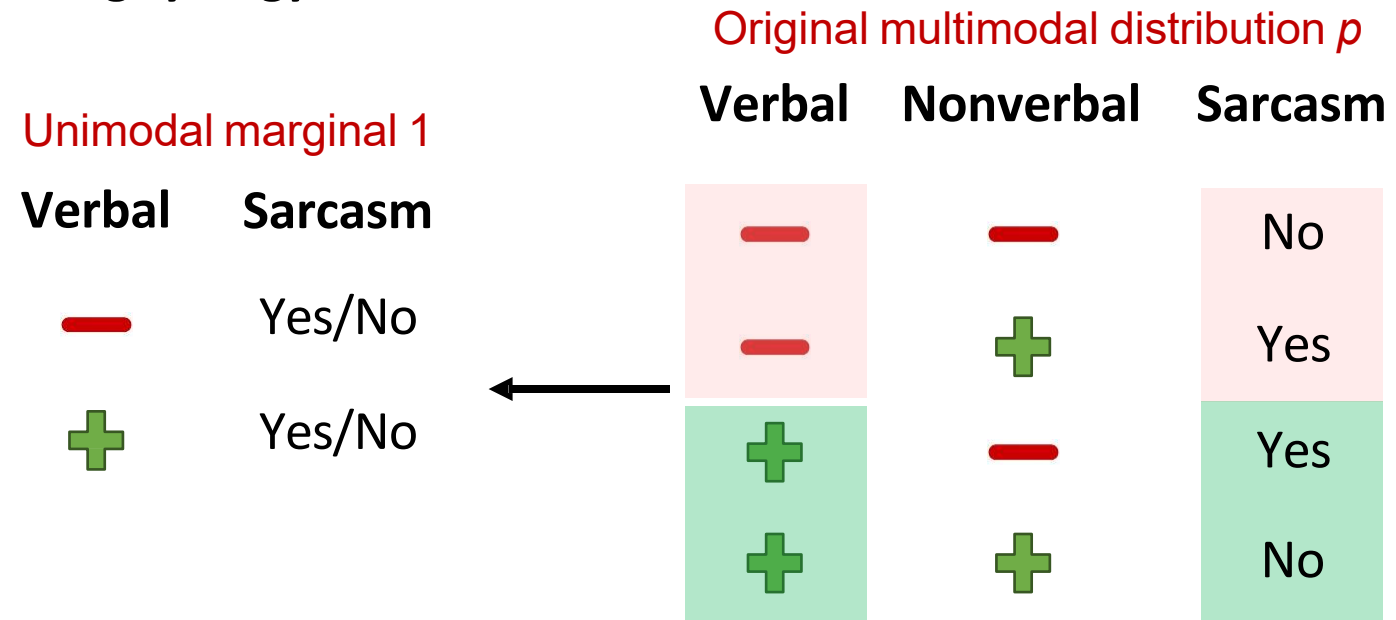
What interactions do models learn?

What models are suitable for my data?



Mathematical Framework for Multimodal Interactions

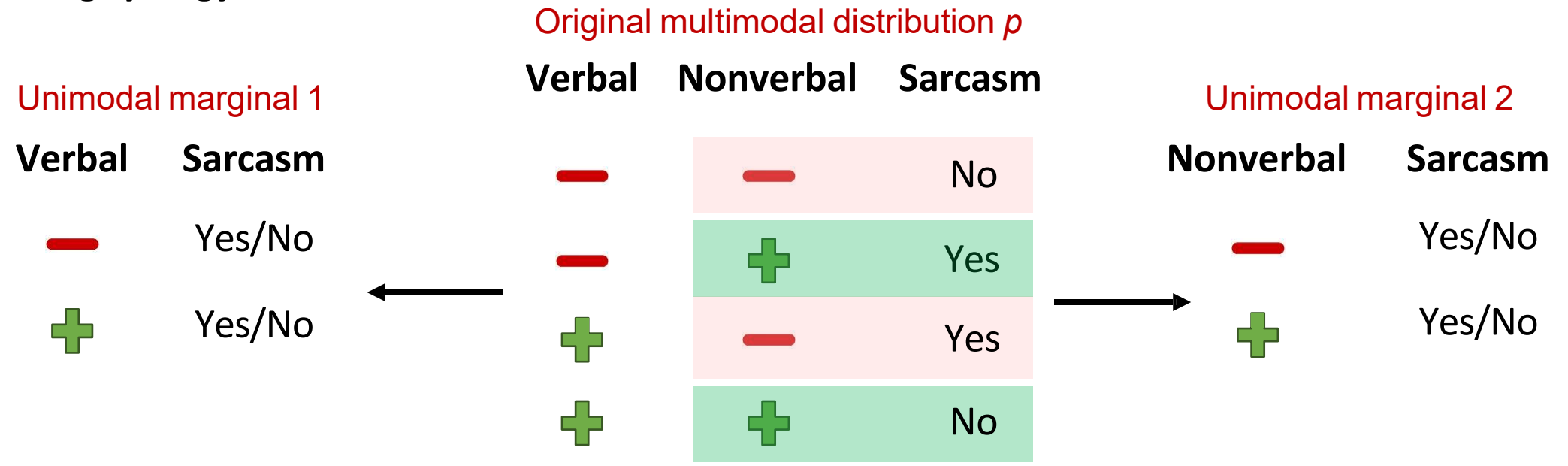
Estimating synergy



Synergy = Original multimodal information about the task
– multimodal information given by the **worst** distribution combining the same modalities

Mathematical Framework for Multimodal Interactions

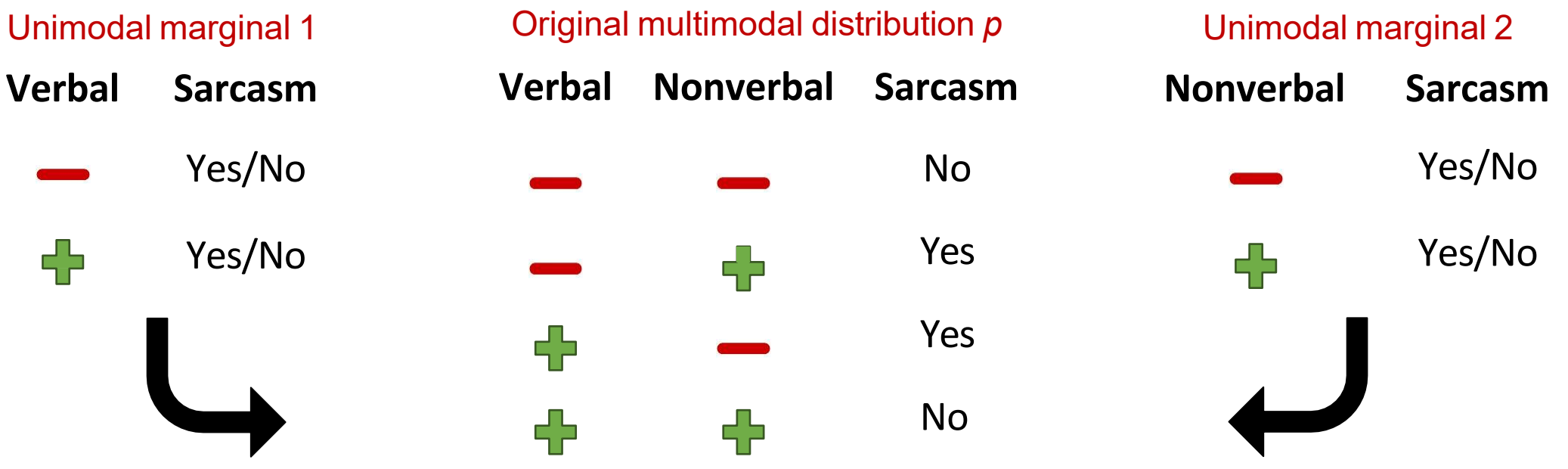
Estimating synergy



Synergy = Original multimodal information about the task
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Mathematical Framework for Multimodal Interactions

Many ways of combining these 2 unimodal marginals into a multimodal distribution!

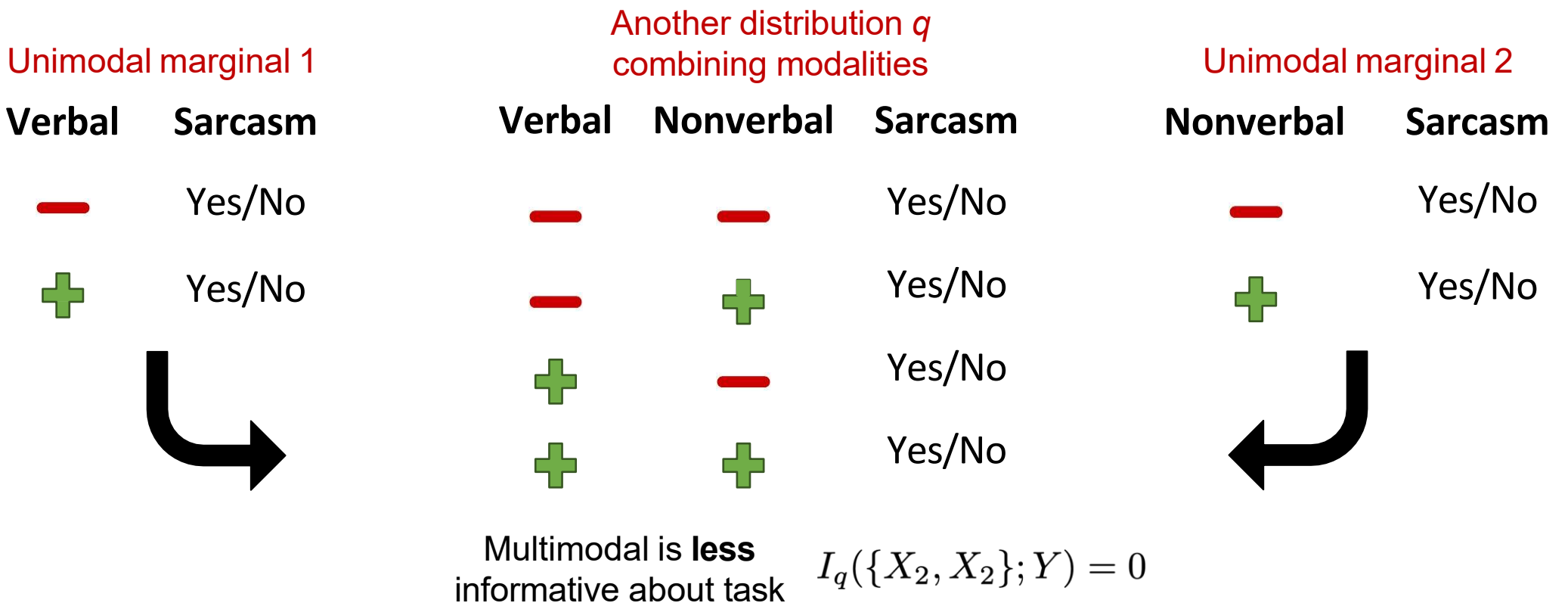


Multimodal is **very** informative about task $I_p(\{X_1, X_2\}; Y) = 1$

Synergy = Original multimodal information about the task
– multimodal information given by the **worst** distribution combining the same modalities

Mathematical Framework for Multimodal Interactions

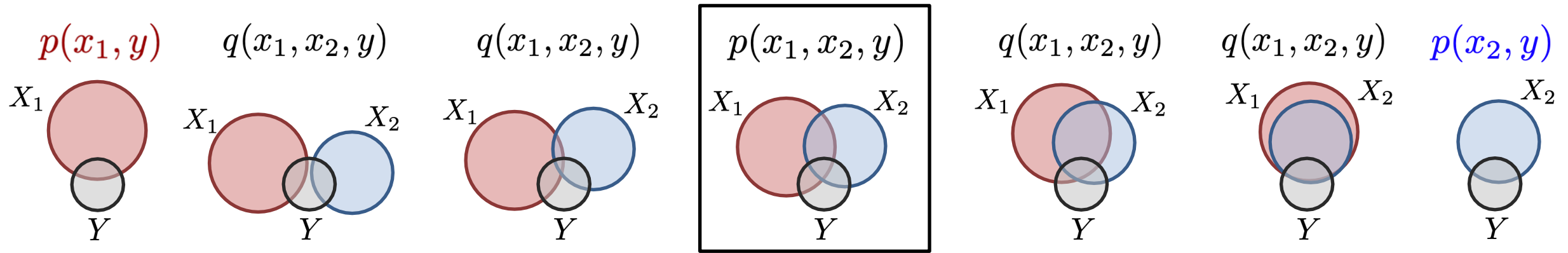
Many ways of combining these 2 unimodal marginals into a multimodal distribution!



Synergy = Original multimodal information about the task
– multimodal information given by the **worst** distribution combining modalities = 1 – 0 = 1

Mathematical Framework for Interactions

More formally as **partial information decomposition**: [Bertschinger et al., 2014]



q must be a **coupling** of the unimodal marginals:

$$\Delta_p = \{q(x_1, x_2, y) : q(x_1, y) = p(x_1, y), q(x_2, y) = p(x_2, y)\}$$

$$S = I_p(X_1, X_2; Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y)$$

Task-relevant
multimodal info

Task-relevant multimodal
info without synergy:

$$S_{q^*} = I_{q^*}(X_1, X_2; Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y) = 0$$

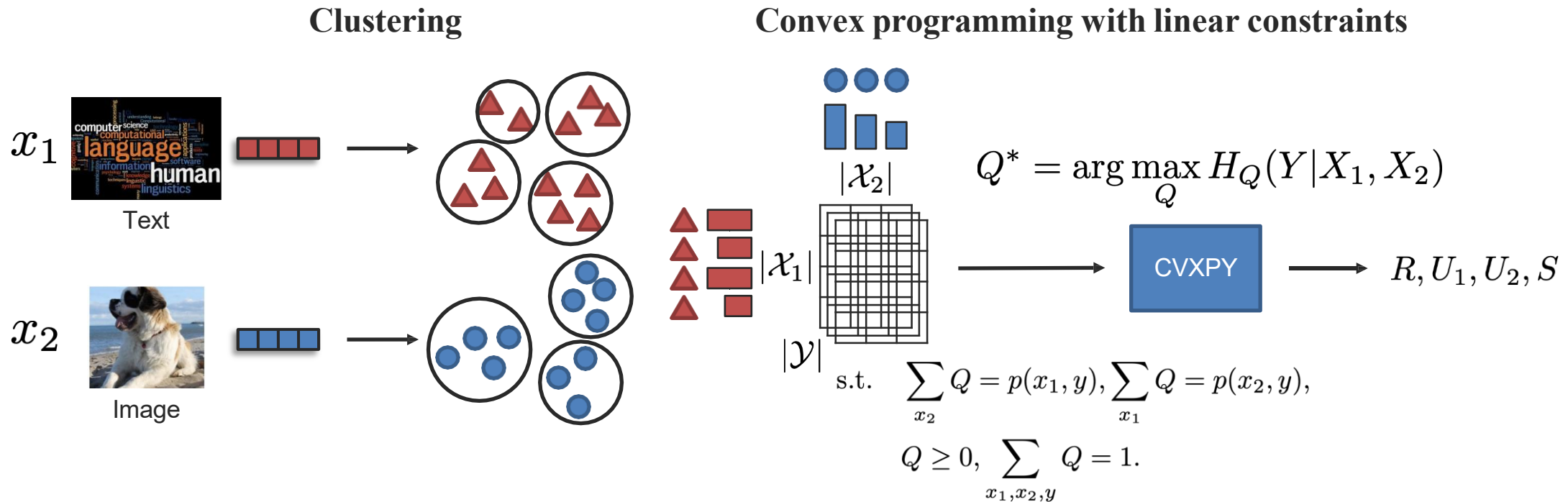
Estimating Partial Information Decomposition

Equivalent formulation as
max-entropy optimization:

$$q^* = \arg \max_{q \in \Delta_p} H_q(Y|X_1, X_2)$$

$$\Delta_p = \{q(x_1, x_2, y) : q(x_1, y) = p(x_1, y), q(x_2, y) = p(x_2, y)\}$$

If X_1, X_2, Y have small discrete support: exact solution via convex programming.



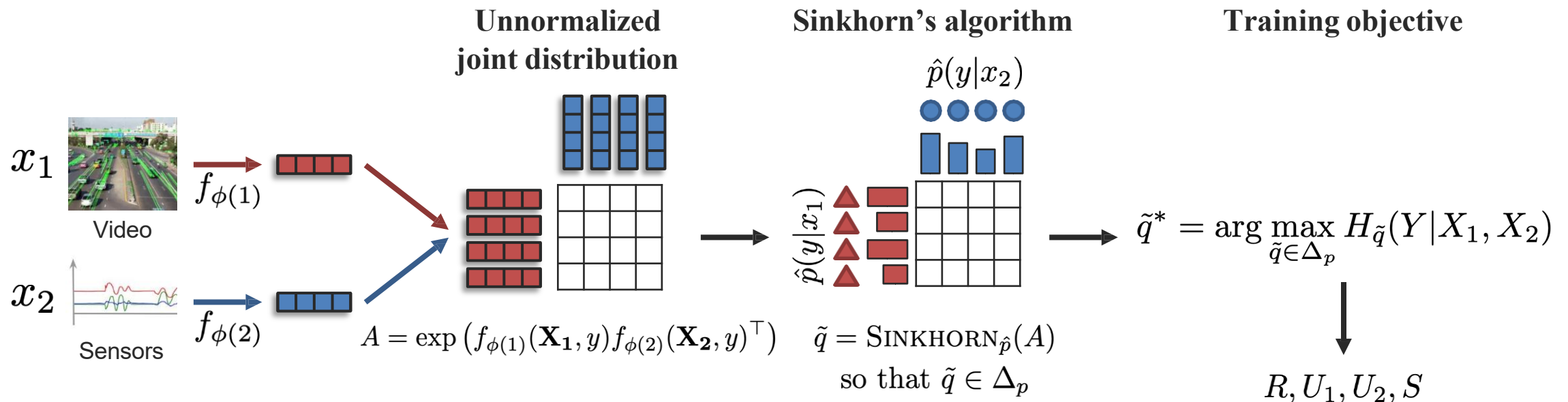
Estimating Partial Information Decomposition

Equivalent formulation as
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$$\Delta_p = \{q(x_1, x_2, y) : q(x_1, y) = p(x_1, y), q(x_2, y) = p(x_2, y)\}$$

If X_1, X_2, Y high-dimensional & continuous: an approximate neural network estimator.



Quantifying Multimodal Datasets

1. Dataset quantification:

$$\mathcal{D} = \{(x_1, x_2, y)\} \longrightarrow \{R, U_1, U_2, S\}_{\mathcal{D}} \bullet$$

Language: *And he I don't think he got mad when hah
I don't know maybe.*

Vision:



Acoustic:

(frustrated voice)

Sentiment



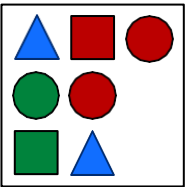
Sheldon :

Its just a *privilege* to watch your
mind at work.

- **Text** : suggests a compliment.
- **Audio** : neutral tone.
- **Video** : straight face.

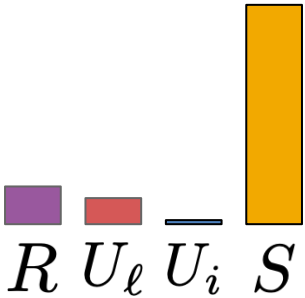


Sarcasm



*Is there a
red shape
above a
circle?*

VQA

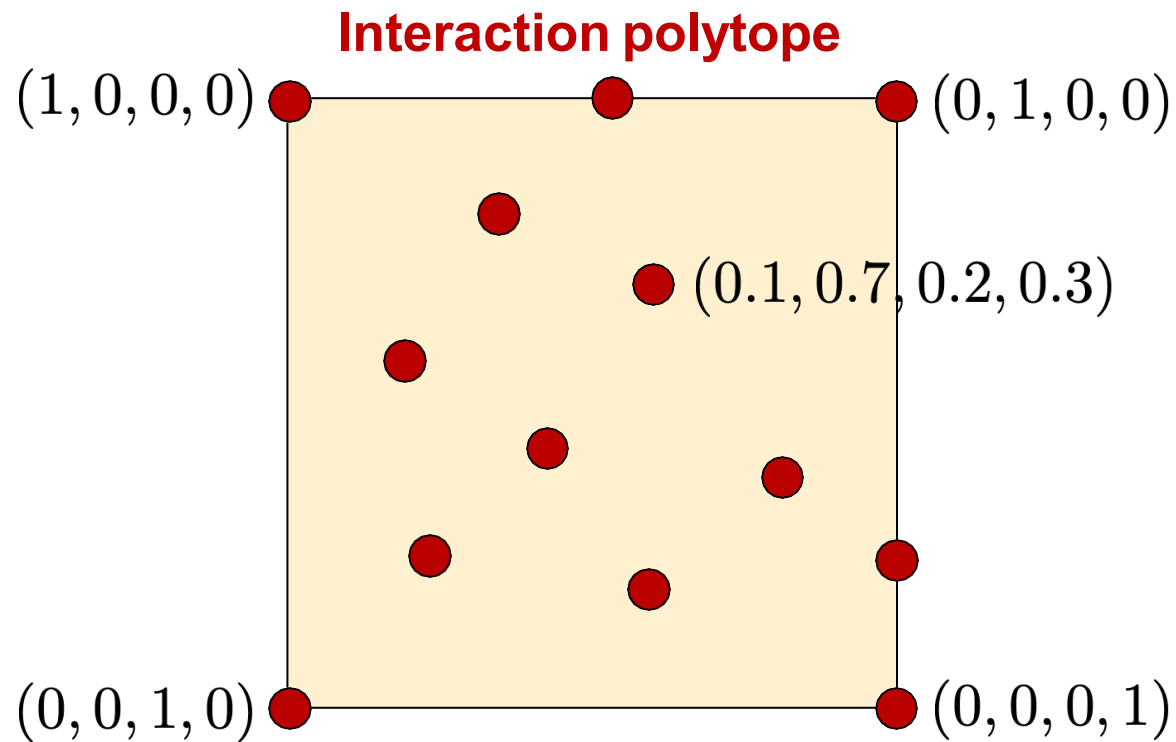


Also matches human judgment of interactions, and other sanity checks on synthetic datasets

Model Selection

1. Dataset quantification:

$$\mathcal{D} = \{(x_1, x_2, y)\} \longrightarrow \{R, U_1, U_2, S\}_{\mathcal{D}} \bullet$$

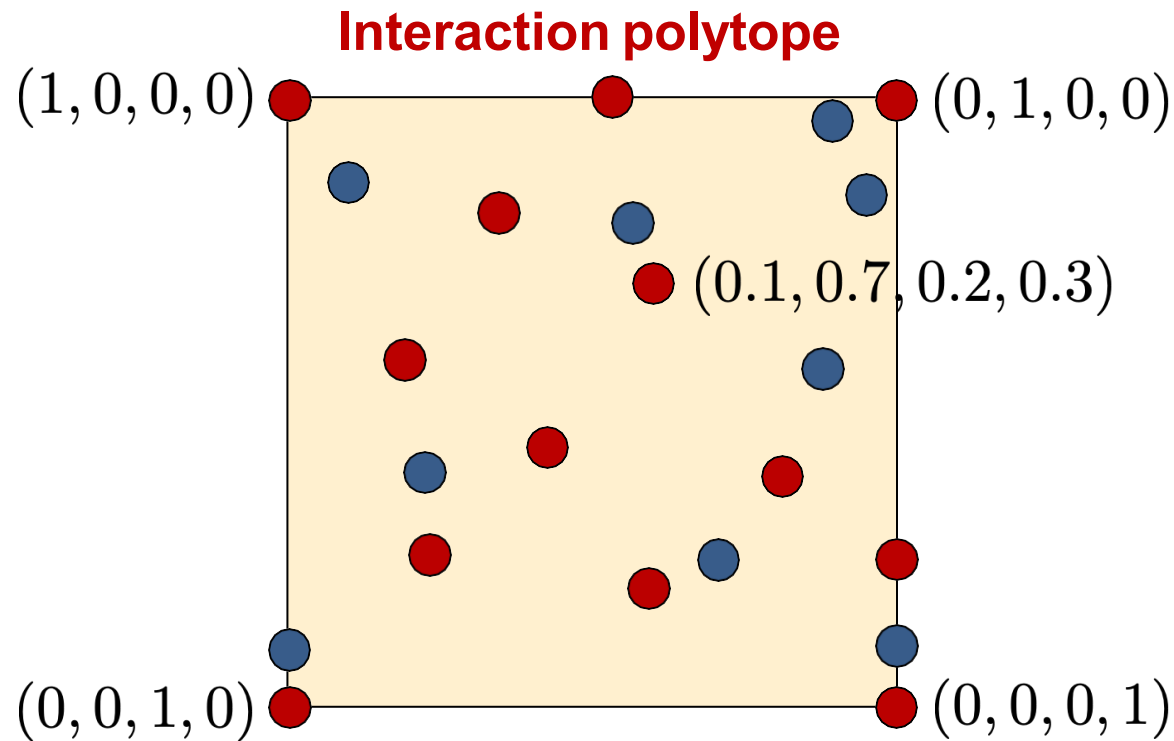


Can be done with
synthetic data

Model Selection

2. Model quantification:

$$\begin{aligned} f(\mathcal{D}) = \{(x_1, x_2, \hat{y} = f(x_1, x_2))\} &\longrightarrow \{R, U_1, U_2, S\}_{f(\mathcal{D})} \\ \{R, U_1, U_2, S\}_{f(\mathcal{D}_1)}, \dots, \{R, U_1, U_2, S\}_{f(\mathcal{D}_k)} &\longrightarrow \{R, U_1, U_2, S\}_f \end{aligned} \quad \bullet$$



Model families trained
on synthetic data

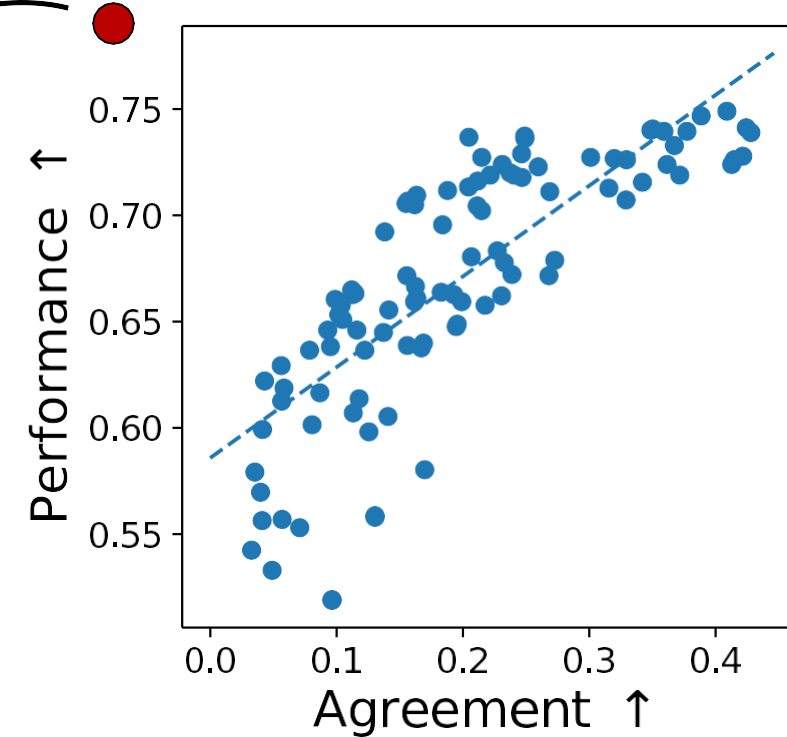
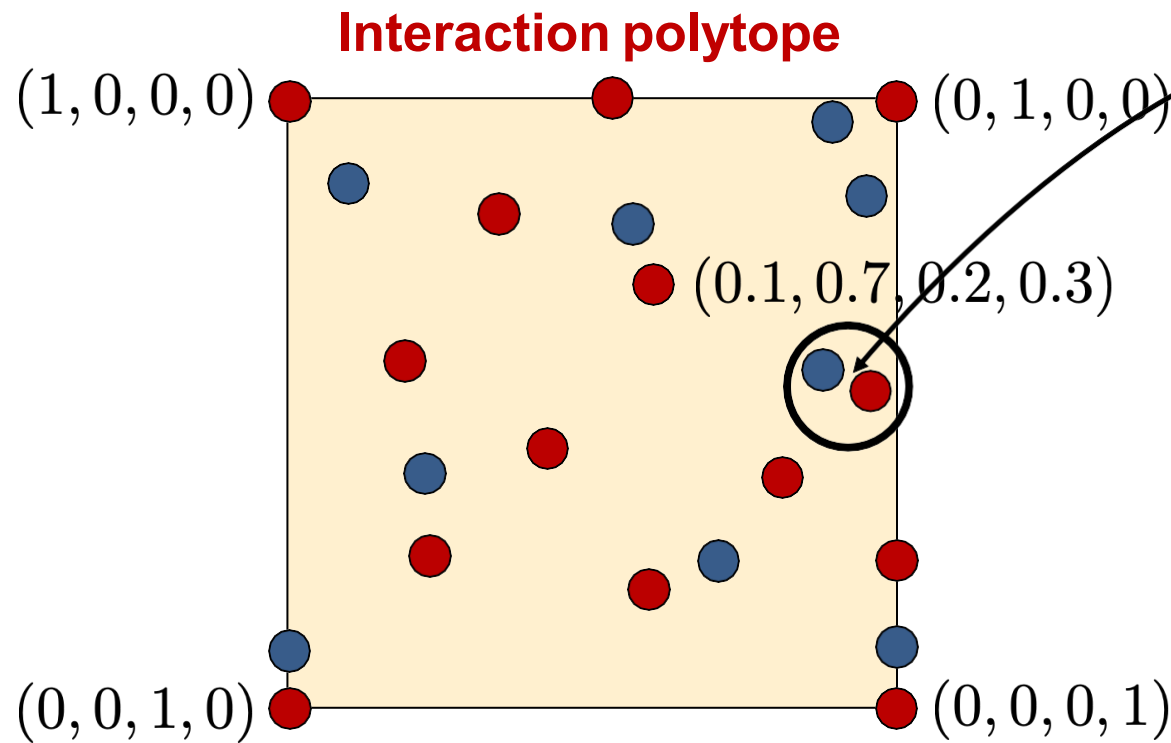
- Unimodal models
- Ensemble
- Multiplicative interactions
- and many more...

Model Selection

3. Model selection:

$$\{R, U_1, U_2, S\}_{\mathcal{D}} \longleftrightarrow \{R, U_1, U_2, S\}_f$$

Selects models with
>96% performance



Model Selection

3. Model selection:

$$\{R, U_1, U_2, S\}_{\mathcal{D}} \longleftrightarrow \{R, U_1, U_2, S\}_f$$

Language: *And he I don't think he got mad when hah
I don't know maybe.*

Vision:



Gaze aversion

Acoustic:

(frustrated voice)

Sentiment



Language/Agreement

Sheldon :

Its just a *privilege* to watch your mind at work.

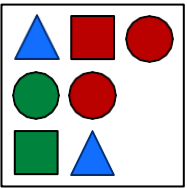
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Sarcasm

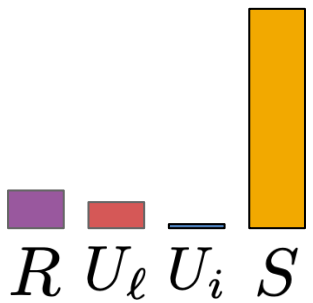


Multimodal Transformer



Is there a red shape above a circle?

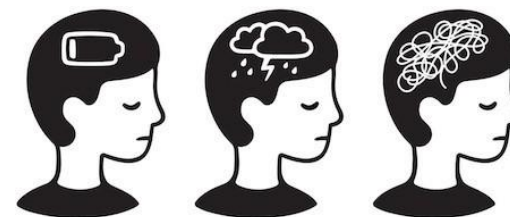
VQA



Multiplicative/Transformer

Application 1: Mental Health

Daily mood prediction as a stepping-stone towards real-time assessment of suicide ideation.



Text + app + keystroke interactions



Slower implies positive	Faster implies positive
just next, was, into, people stuff, cute, phone, want, talk, see don't, talk	why, thank, haha making, work, idk they, send, dont, man, going think, you, all, love

+ words like 'love', 'thanks', 'haha' become more positive when typed faster

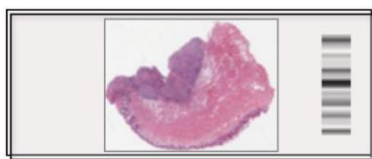
- words like 'don't', 'just' become more negative when typed faster

$$R \ U_{\ell} \ U_{ak} \ S$$

Application 2: Computational Pathology



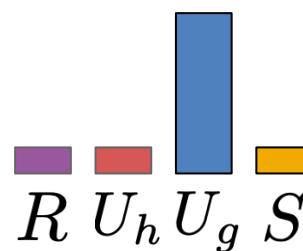
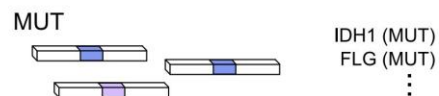
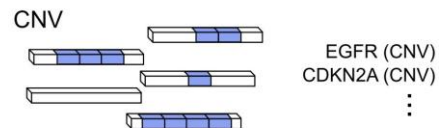
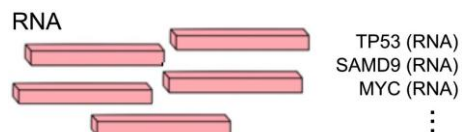
Histology images



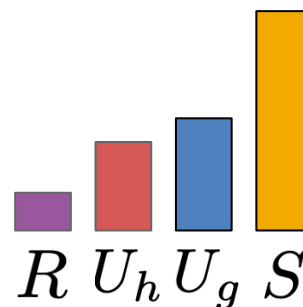
⋮



Genomics profile



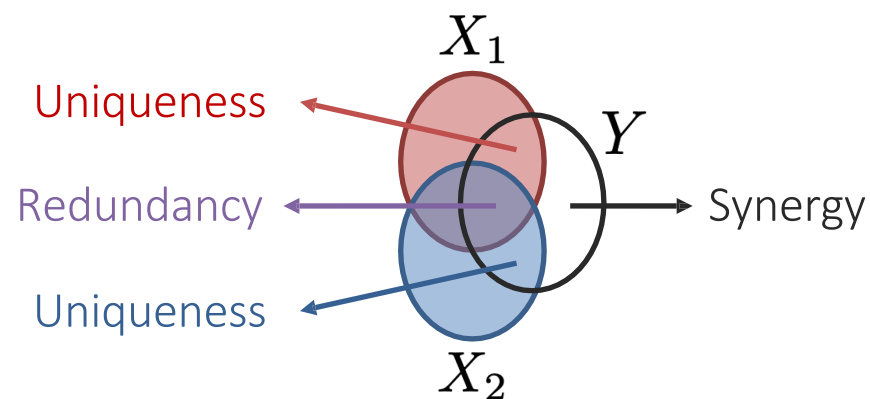
Glioma: Genomics unimodal



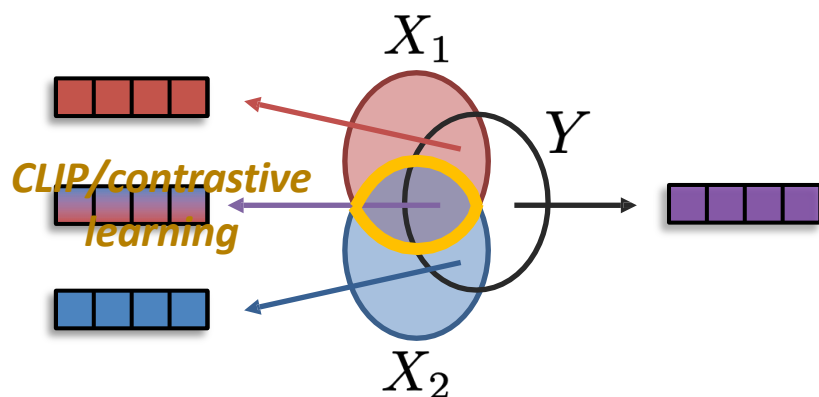
Pancreas: Histology + genomics interaction

Understanding the models and adoption in practice by doctors

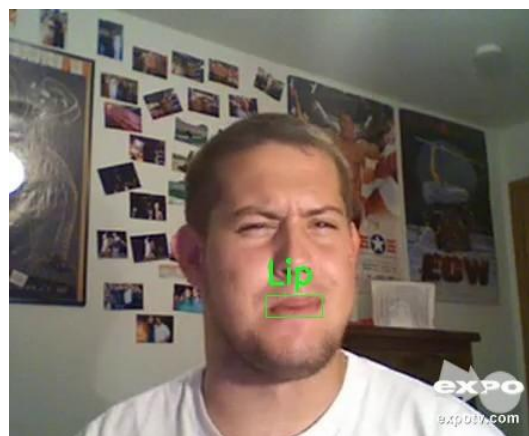
Implications of Studying Multimodal Interactions



Optimizing these interactions as training objectives:



Visualizing the interactions learned in individual neurons:



Why am I spending my money watching this? *(sigh)* I think I was more *sad*...

Predicting multimodal performance to decide modality utility:

$$p(x_1, y) \quad f_1: \blacktriangle \longrightarrow y_1$$

$$p(x_2, y) \quad f_2: \bullet \longrightarrow y_2$$



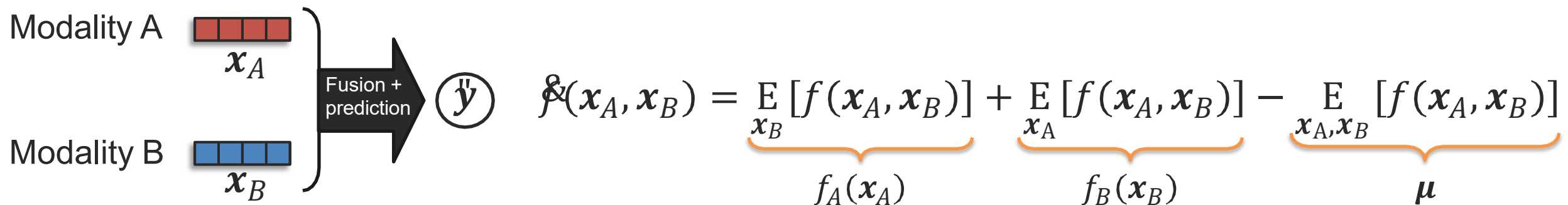
$$p(x_1, x_2, y) \quad \text{80\% accuracy}$$

Quantifying Cross-modal Interactions

Identifying overall presence of cross-modal interactions

Statistical non-additive interactions [Friedman & Popescu, 2008, Sorokina et al., 2008]

f exhibits interactions between 2 features x_A and x_B iff f cannot be decomposed into a sum of unimodal subfunctions f_A, f_B such that $f(x_A, x_B) = f_A(x_A) + f_B(x_B)$.



If the additive projection $\mathcal{F}(x_A, x_B)$ is equal to nonlinear fusion $f(x_A, x_B)$ then the non-additive interactions are not modeled.

μ measures **overall quantity** of cross-modal interactions on a trained model + dataset.

Quantifying Cross-modal Interactions

Identifying individual cross-modal interactions

Statistical non-additive interactions [Friedman & Popescu, 2008, Sorokina et al., 2008]

f exhibits interactions between 2 features x_A and x_B iff f cannot be decomposed into a sum of unimodal subfunctions f_A, f_B such that $f(x_A, x_B) = f_A(x_A) + f_B(x_B)$.

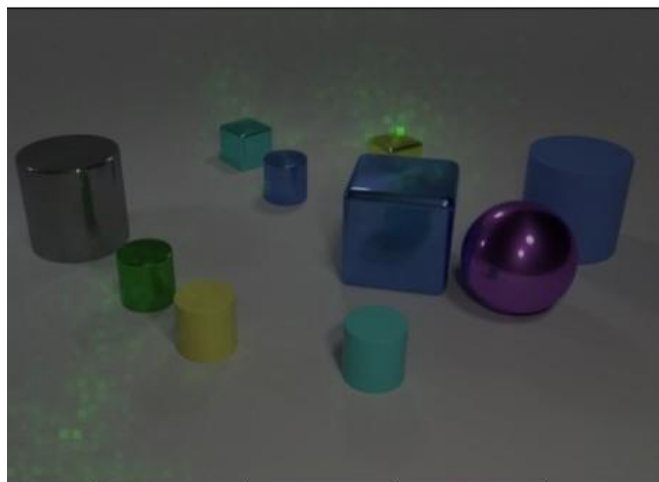
f exhibits interactions between 2 features x_A and x_B iff $\frac{\partial^2 f}{\partial x_A \partial x_B} > 0$.

Natural second-order extension of gradient-based approaches!

Quantifying Cross-modal Interactions

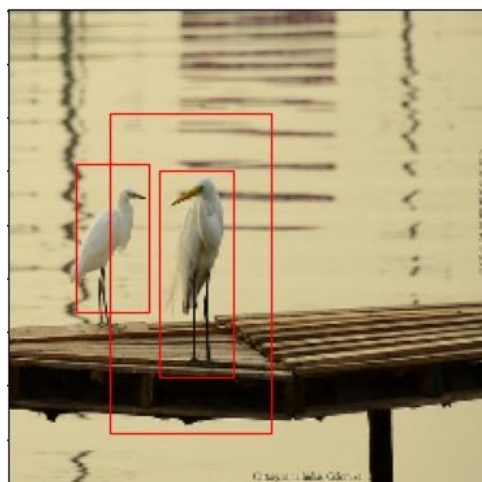
Identifying individual cross-modal interactions

CLEVR



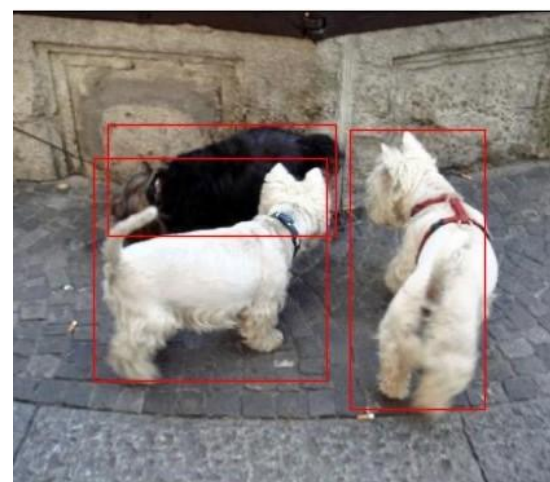
The other small shiny thing that is the same shape as the **tiny yellow shiny object** is what color?

VQA 2.0



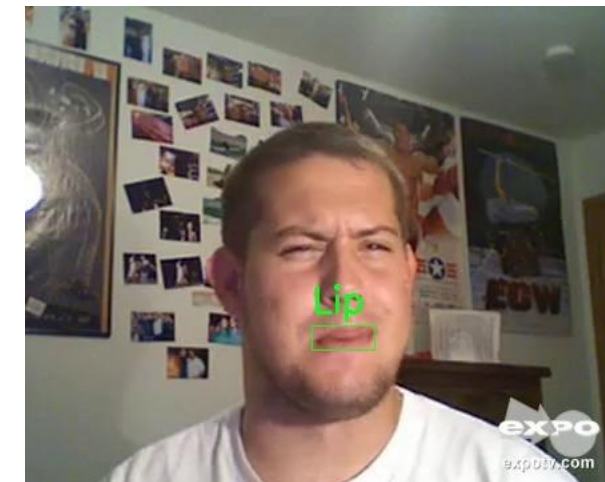
How many **birds**?

Flickr-30k



Three small dogs, two white and one black and white, on a sidewalk.

CMU-MOSEI



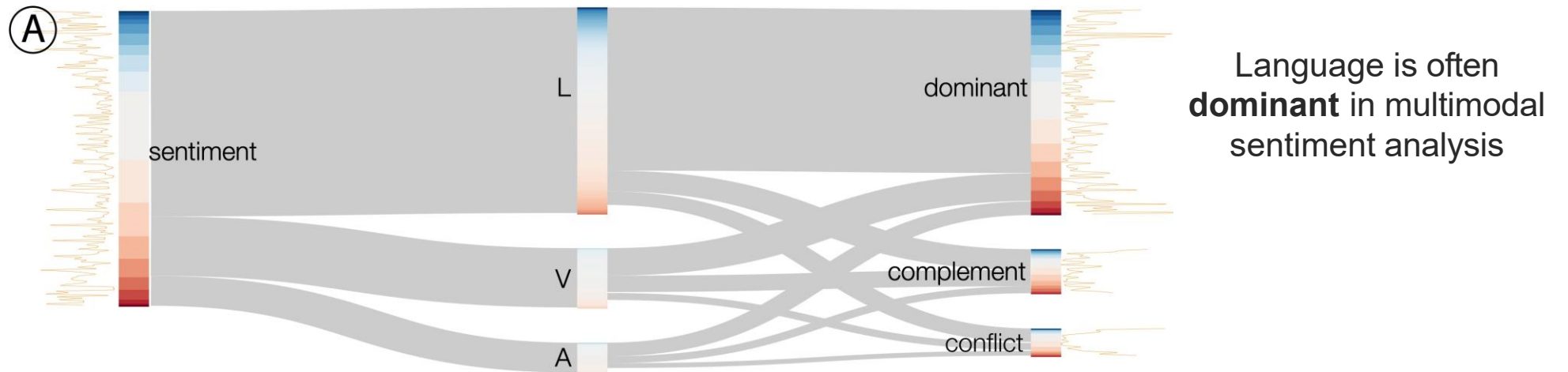
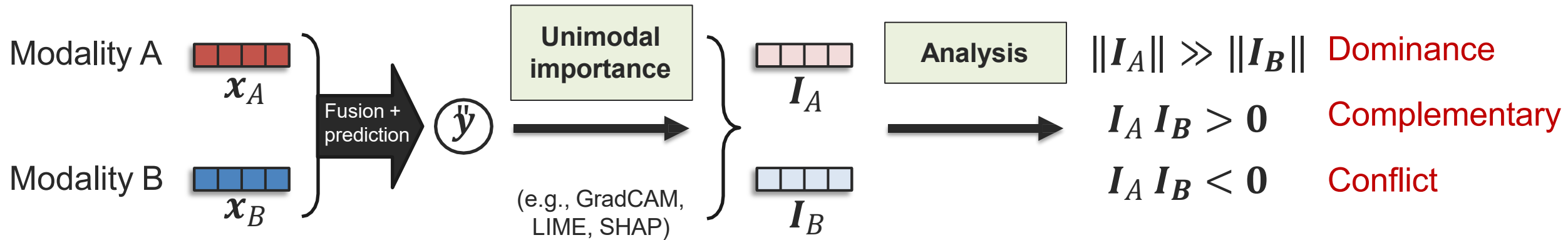
Why am I spending my money watching this? (**sigh**) I think I was more **sad**...

Correspondence

Relationships

Quantifying Cross-modal Interactions

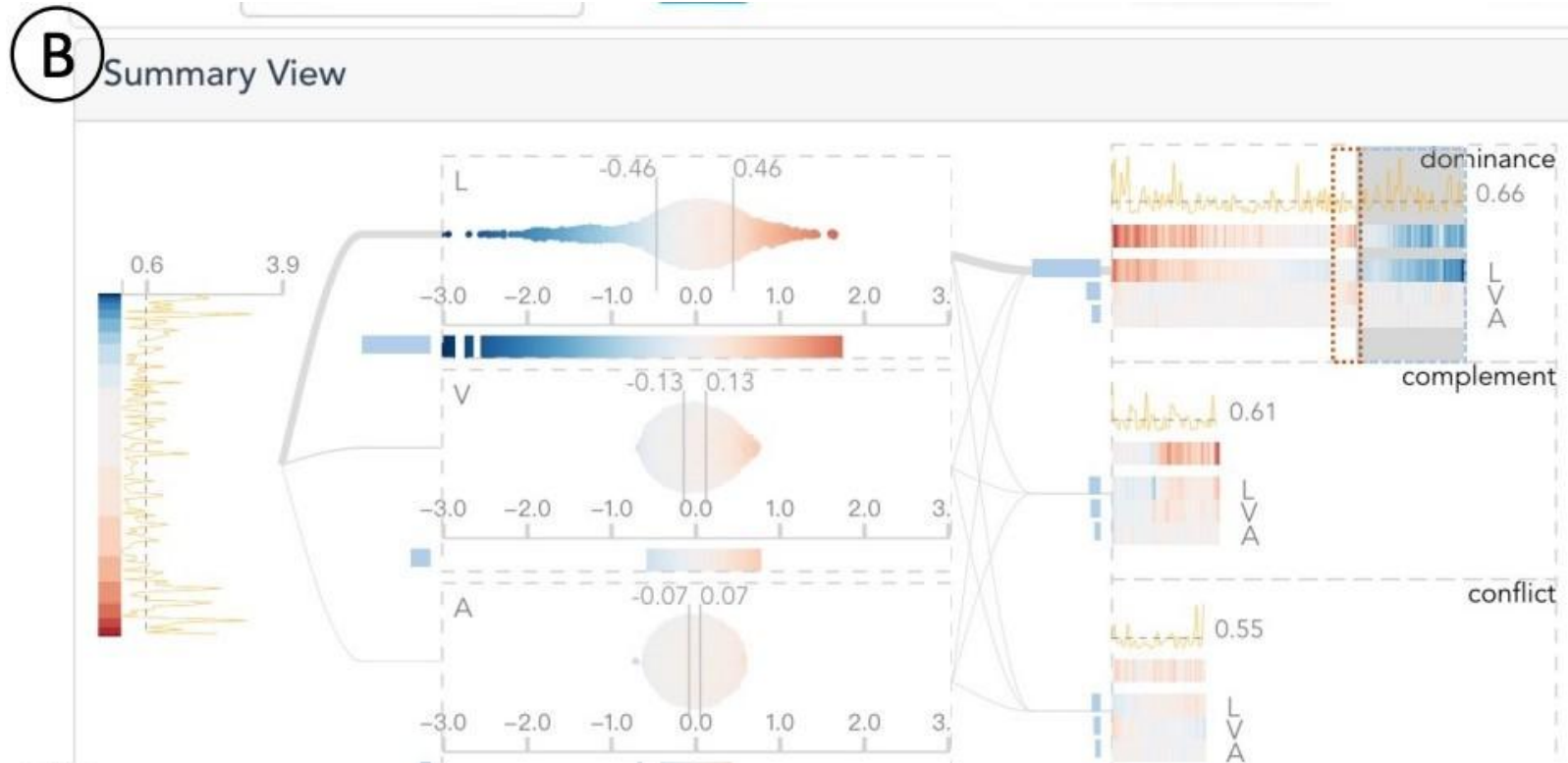
Classification of cross-modal interactions



Quantifying Cross-modal Interactions

Visualization website

See interactive website: <https://andy-xingbowang.com/m2lens/>



Summary of
cross-modal interactions
across entire dataset.

Quantifying Cross-modal Interactions

Visualization website

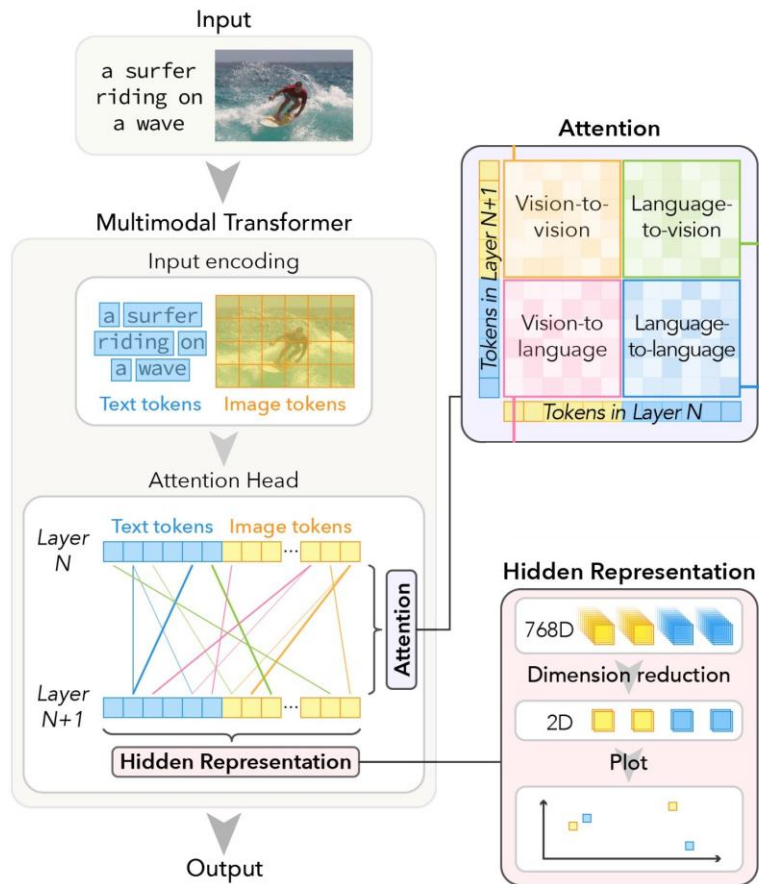
See interactive website: <https://andy-xingbowang.com/m2lens/>



Summary of
cross-modal interactions
in a single instance.

Quantifying Cross-modal Interactions

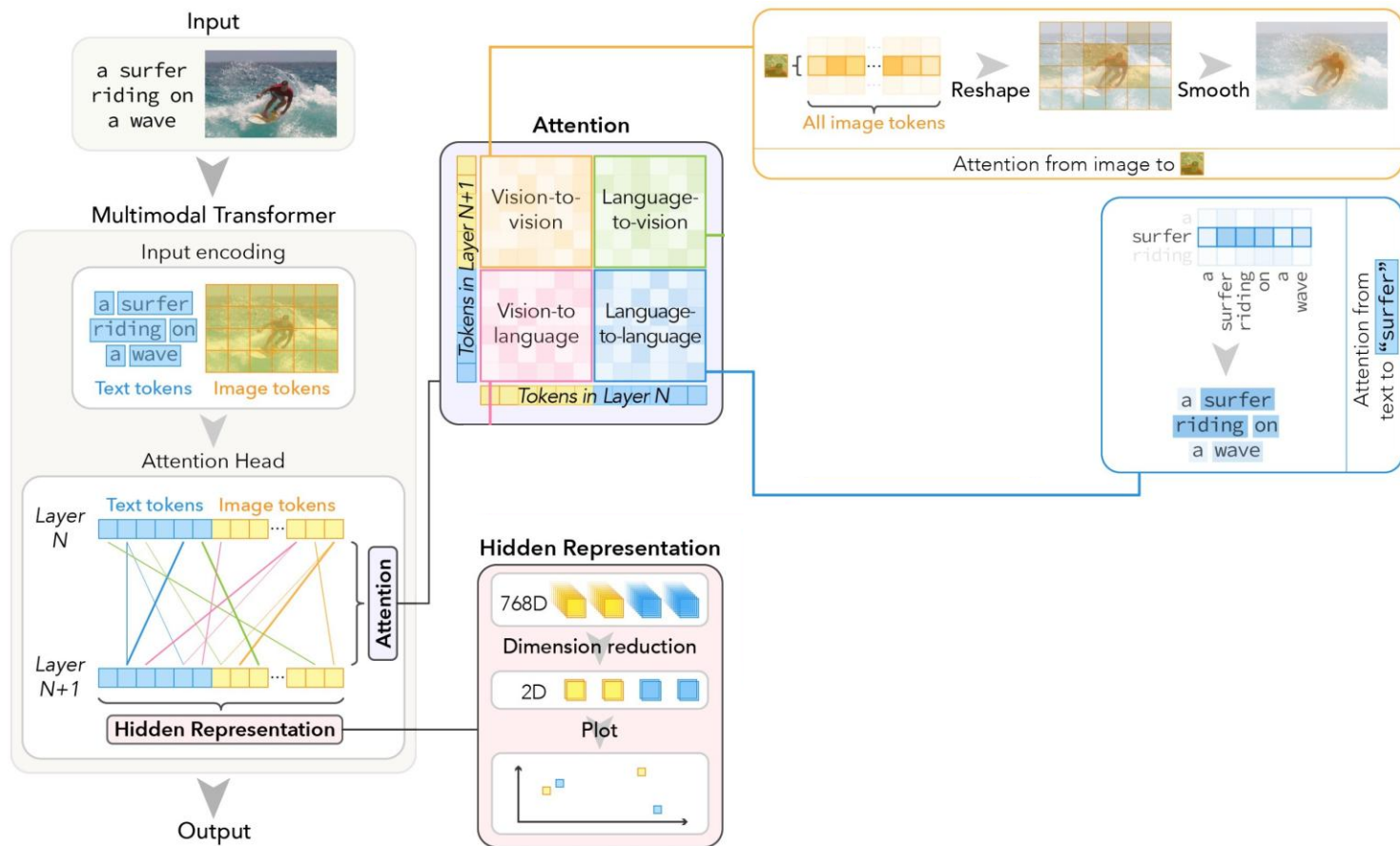
Visualizing multimodal transformers See interactive website: <https://github.com/IntelLabs/VL-InterpreT>



[Aflalo et al., VL-InterpreT: An Interactive Visualization Tool for Interpreting Vision-Language Transformers. CVPR 2022]

Quantifying Cross-modal Interactions

Visualizing multimodal transformers See interactive website: <https://github.com/IntelLabs/VL-InterpreT>

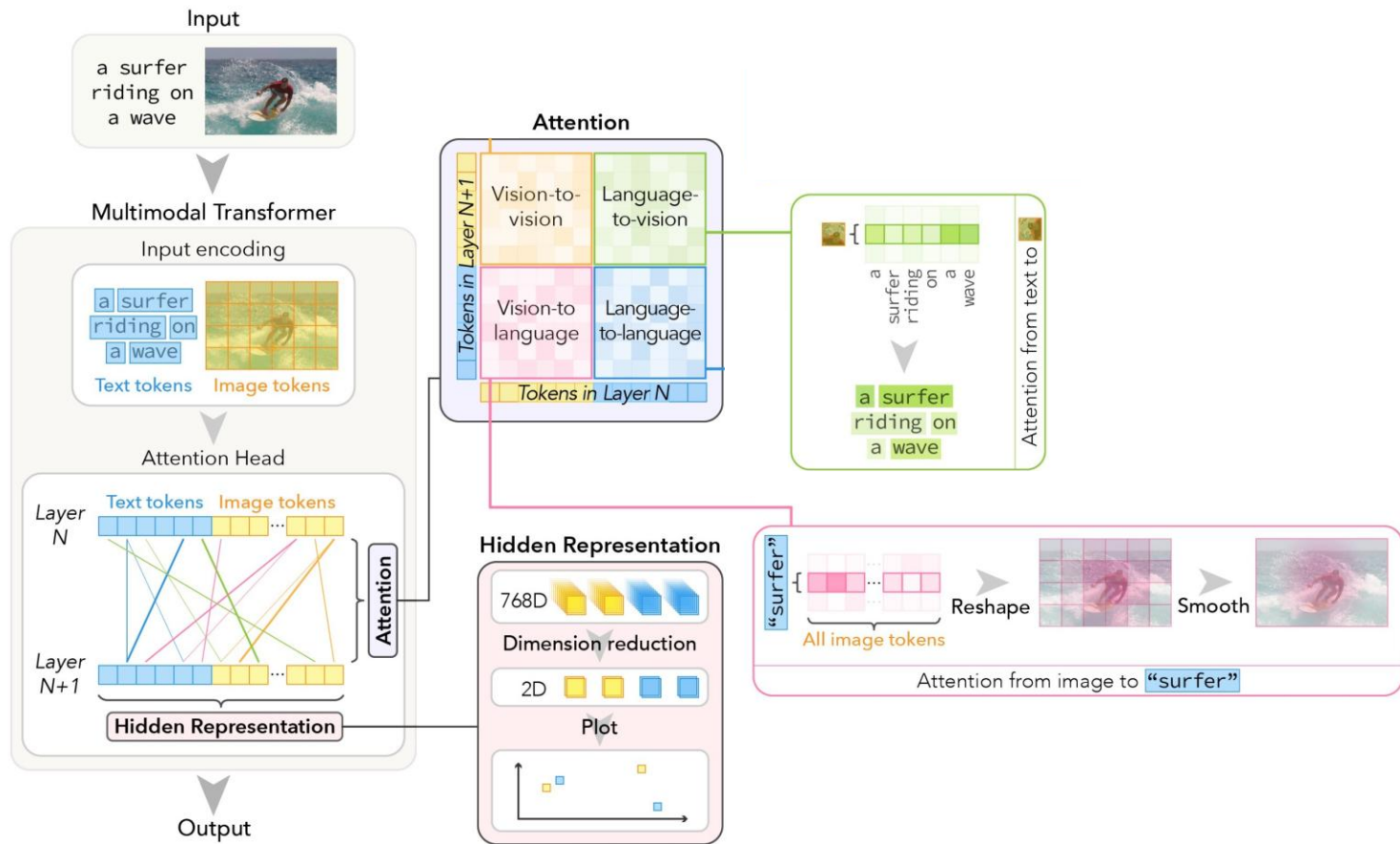


Unimodal image importance

Unimodal text importance

Quantifying Cross-modal Interactions

Visualizing multimodal transformers See interactive website: <https://github.com/IntelLabs/VL-InterpreT>

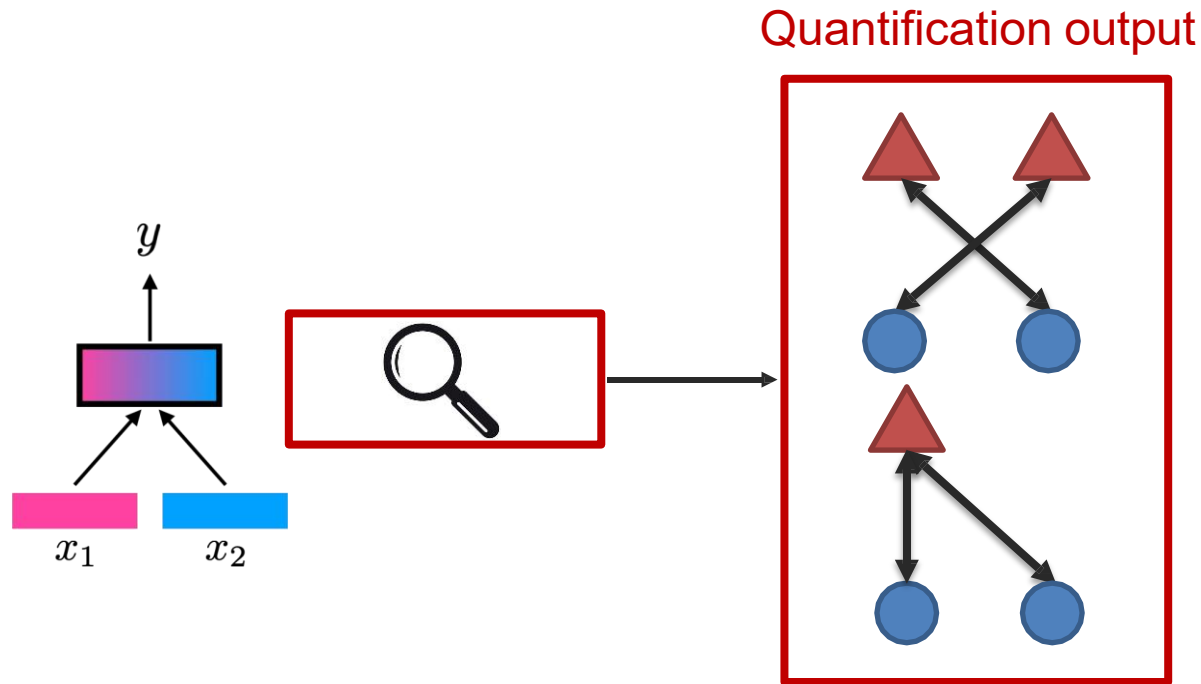


Correspondence and complementary interactions

Evaluating Quantification

How can we evaluate the success of quantifying cross-modal interactions?

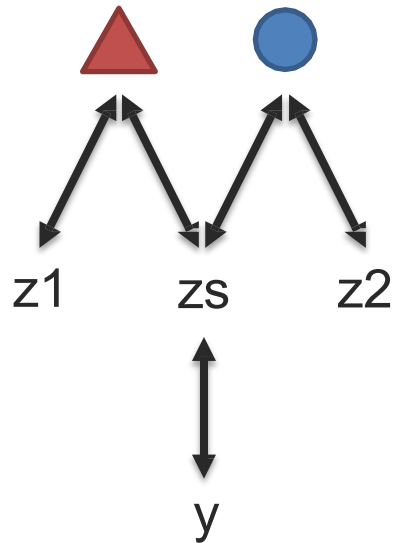
Problem: real-world datasets and models do not have cross-modal interactions annotated!



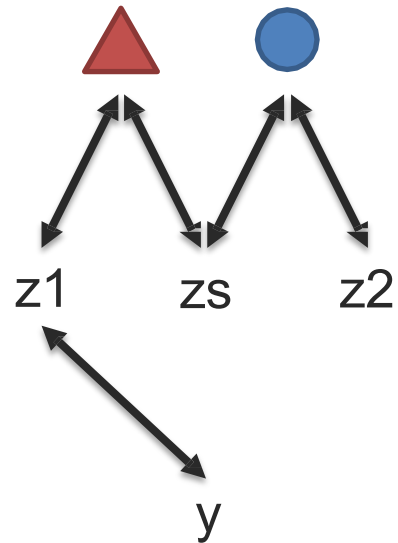
Directly Evaluating Quantification

Direct evaluation: Create datasets for each tested quality, but limited to synthetic data

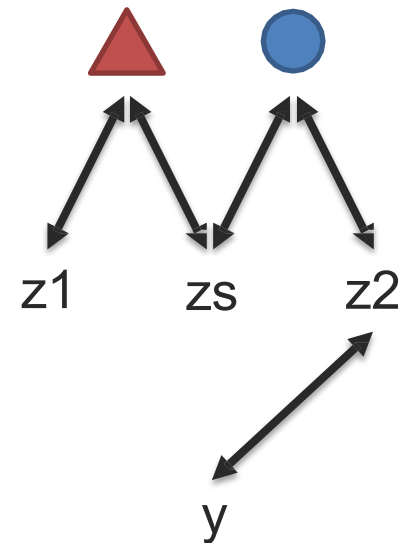
$$\mathcal{D} = \{(x_1, x_2, y)\} \longrightarrow \{R, U_1, U_2, S\}_{\mathcal{D}}$$



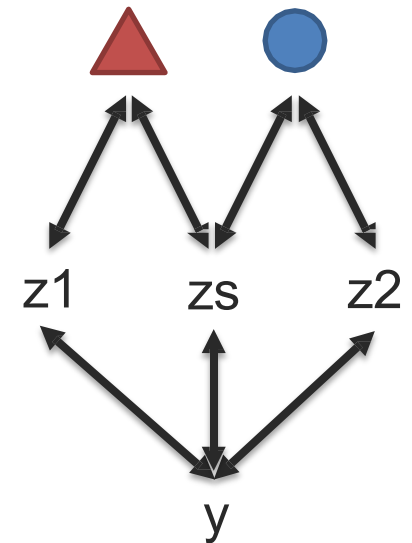
Redundancy



Unique 1



Unique 2



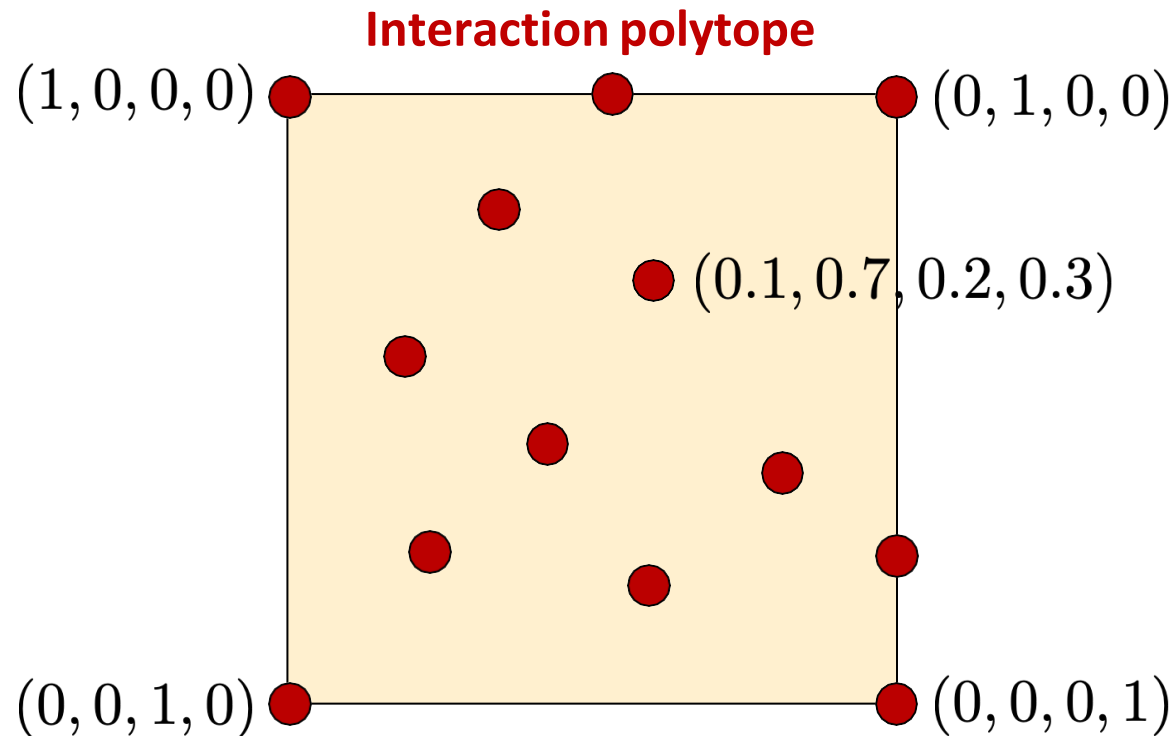
Synergy

Directly Evaluating Quantification

Direct evaluation: Create datasets for each tested quality, but limited to synthetic data

$$\mathcal{D} = \{(x_1, x_2, y)\} \longrightarrow \{R, U_1, U_2, S\}_{\mathcal{D}}$$

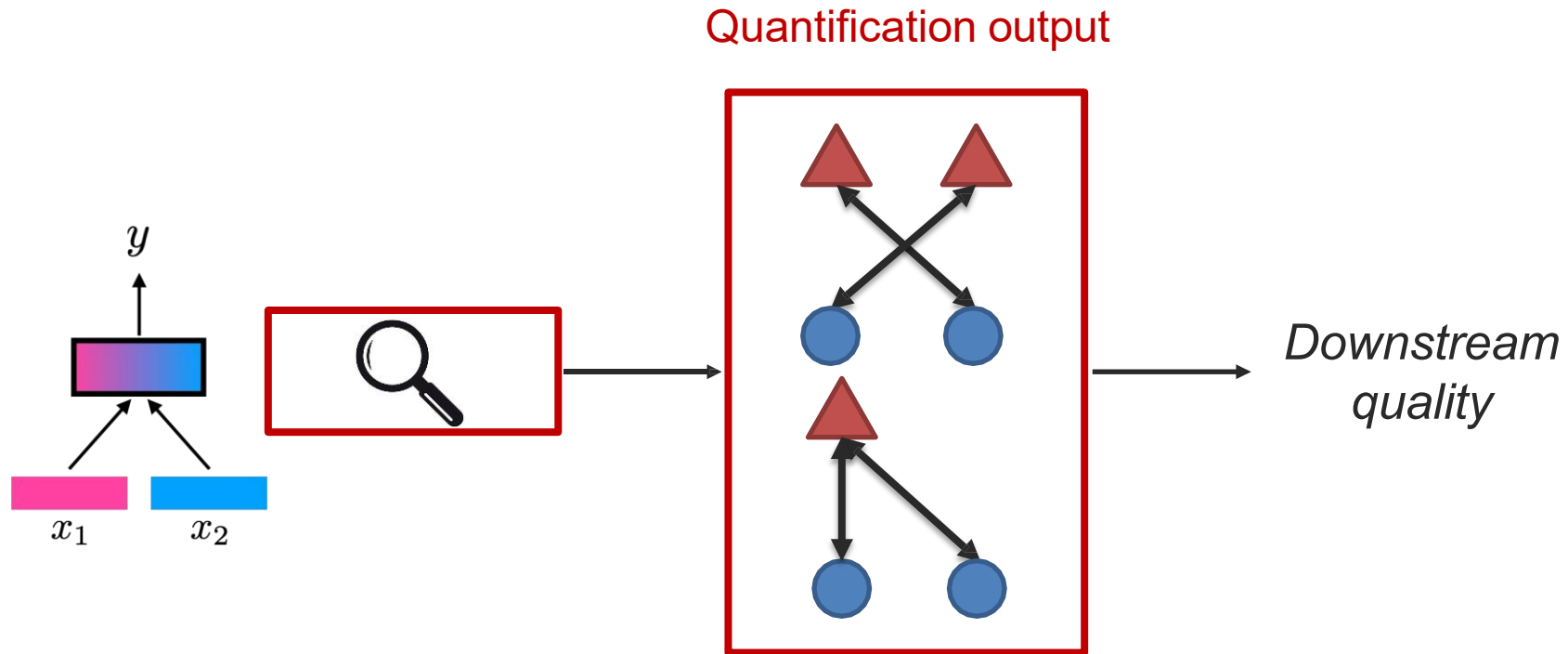
Can be done with
synthetic data



Indirectly Evaluating Quantification

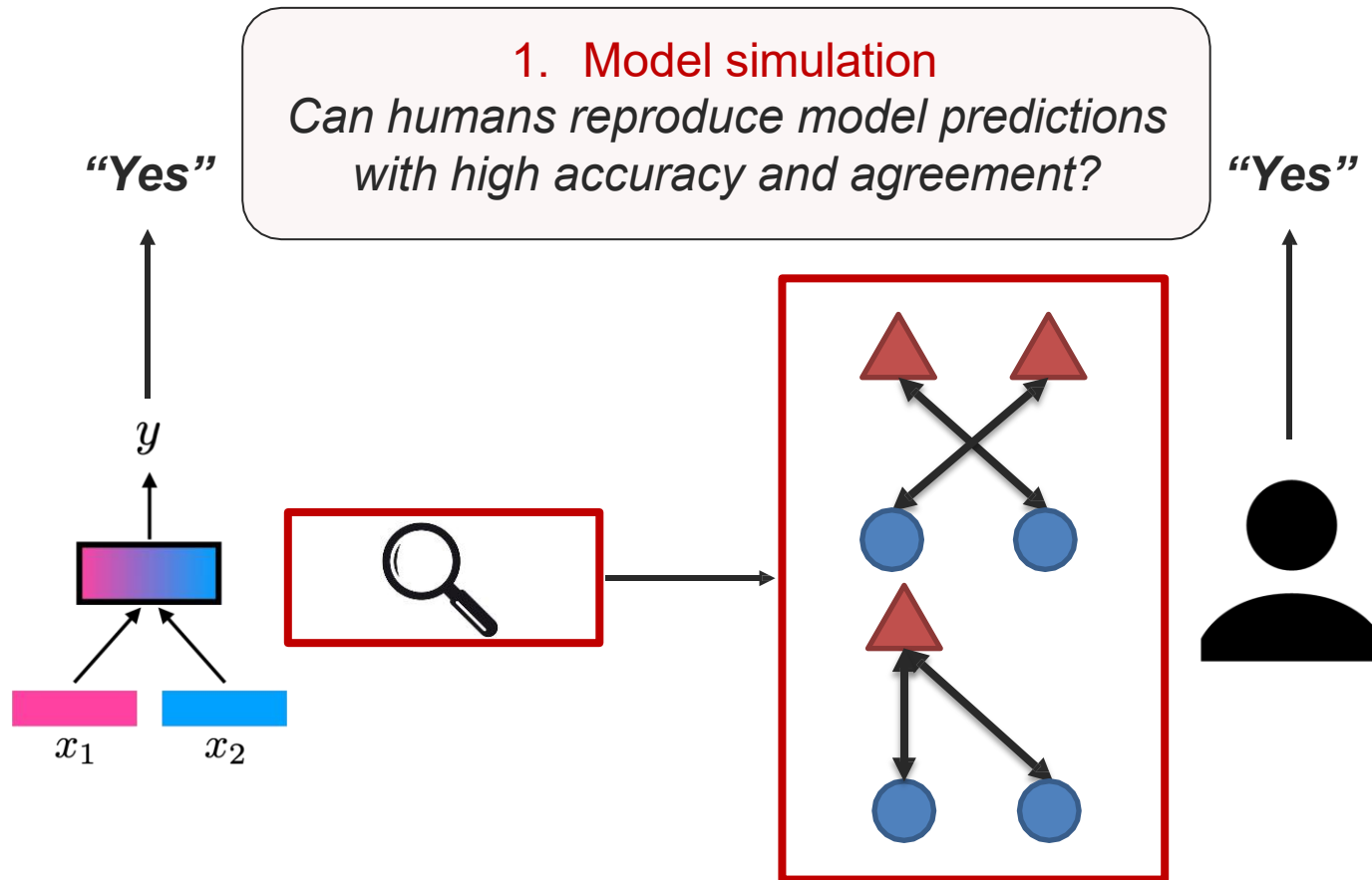
Indirect evaluation

*Find some downstream quality that practitioners find useful
and can be easily evaluated.*



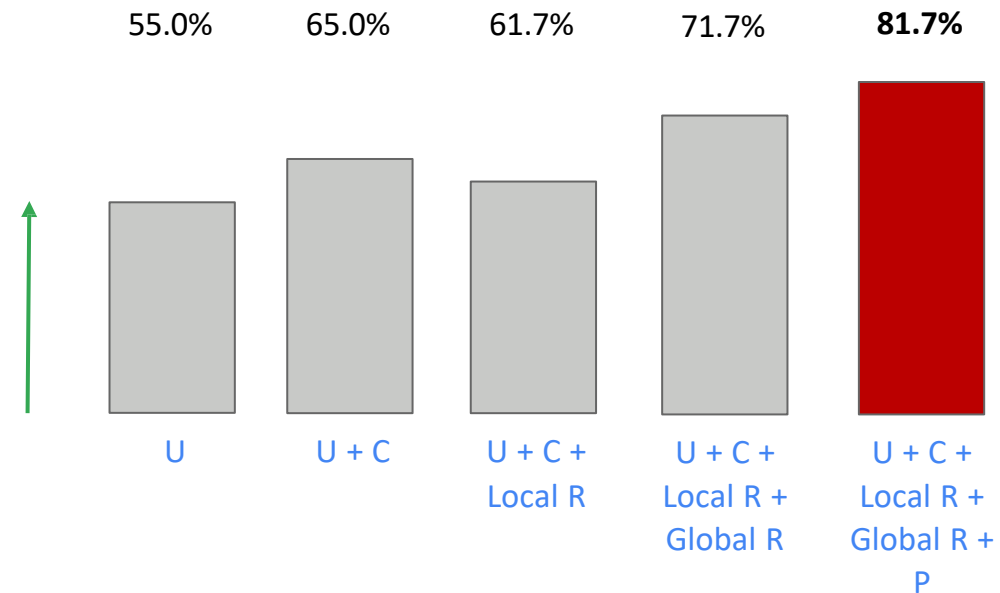
Indirectly Evaluating Quantification

Indirect evaluation: Model simulation



Indirectly Evaluating Quantification

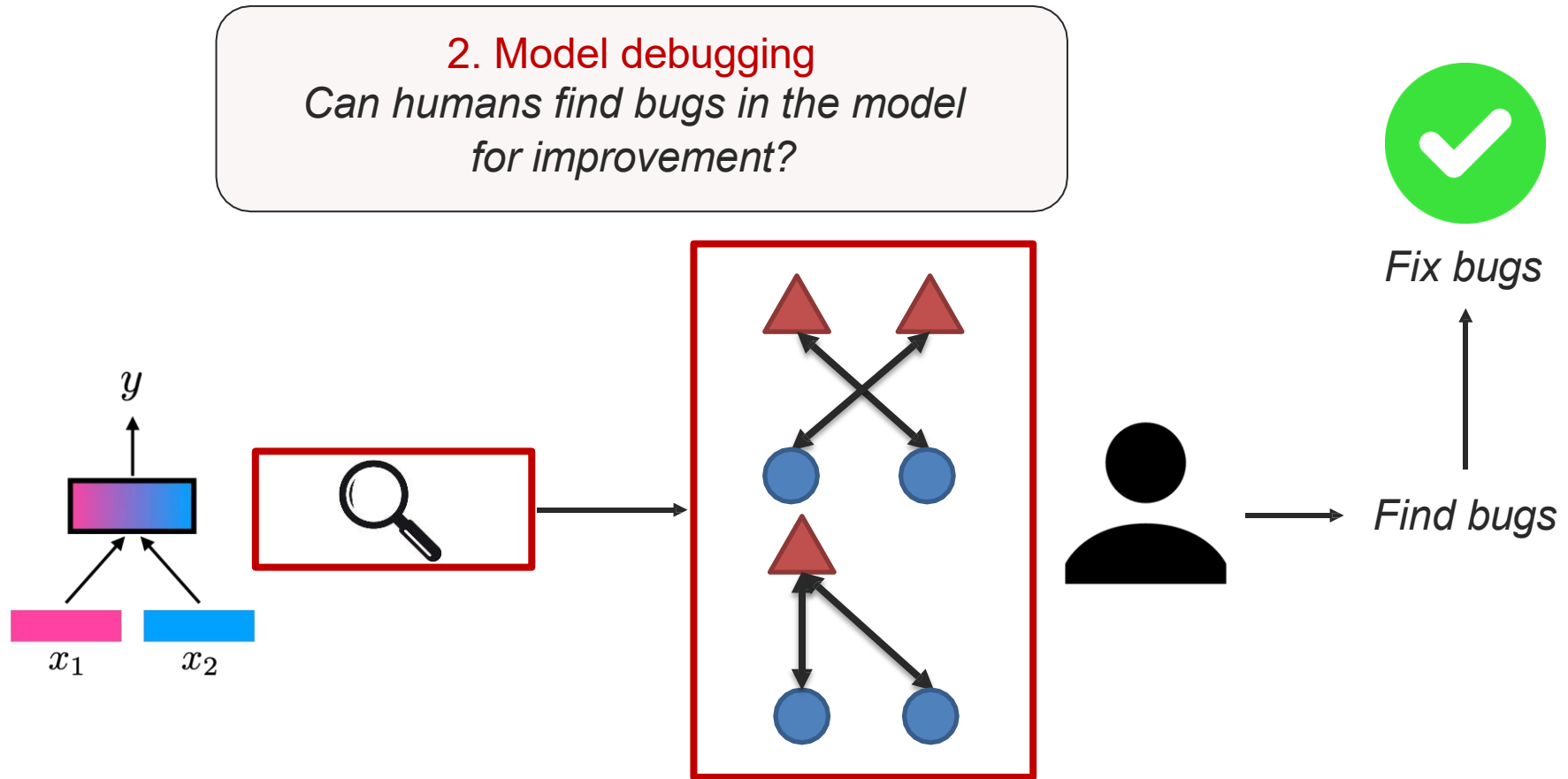
Indirect evaluation: Model simulation



MultiViz stages leads to higher accuracy and agreement
Blind test + reasonable baselines + measurable outcome

Indirectly Evaluating Quantification

Indirect evaluation: Model error analysis and debugging



Indirectly Evaluating Quantification

Indirect evaluation: Model error analysis and debugging



What color is the tie of the second man to the left?



What color is the Salisbury Rd sign?



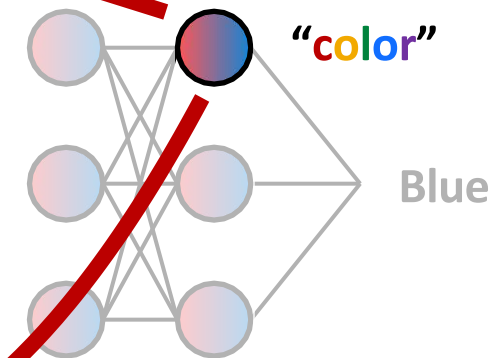
What color is the building?



What color are the checkers on the wall?

Local analysis

3. Multimodal representations



Global analysis

“Models pick up cross-modal interactions but fail in identifying color!”

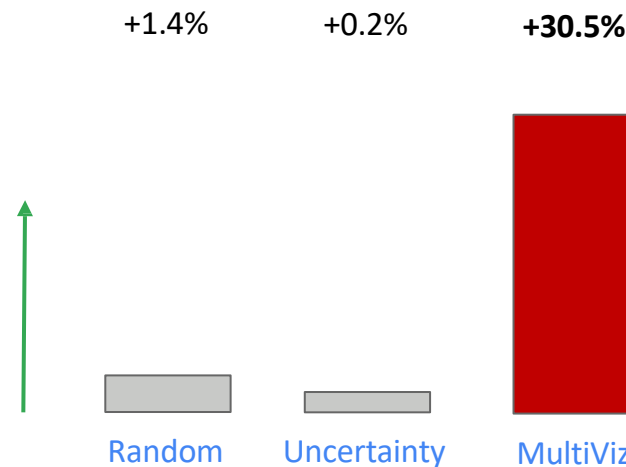
Indirectly Evaluating Quantification

Indirect evaluation: Model error analysis and debugging

“Models pick up cross-modal interactions but fail in identifying color!”



Add targeted examples involving color.



Side note: we used this to discover a bug in a popular deep learning code repository.


Transformers

MultiViz enables error analysis and debugging of multimodal models

[Liang et al., MultiViz: Towards Visualizing and Understanding Multimodal Models. ICLR 2023]

Indirectly Evaluating Quantification

More ways for indirect evaluation:

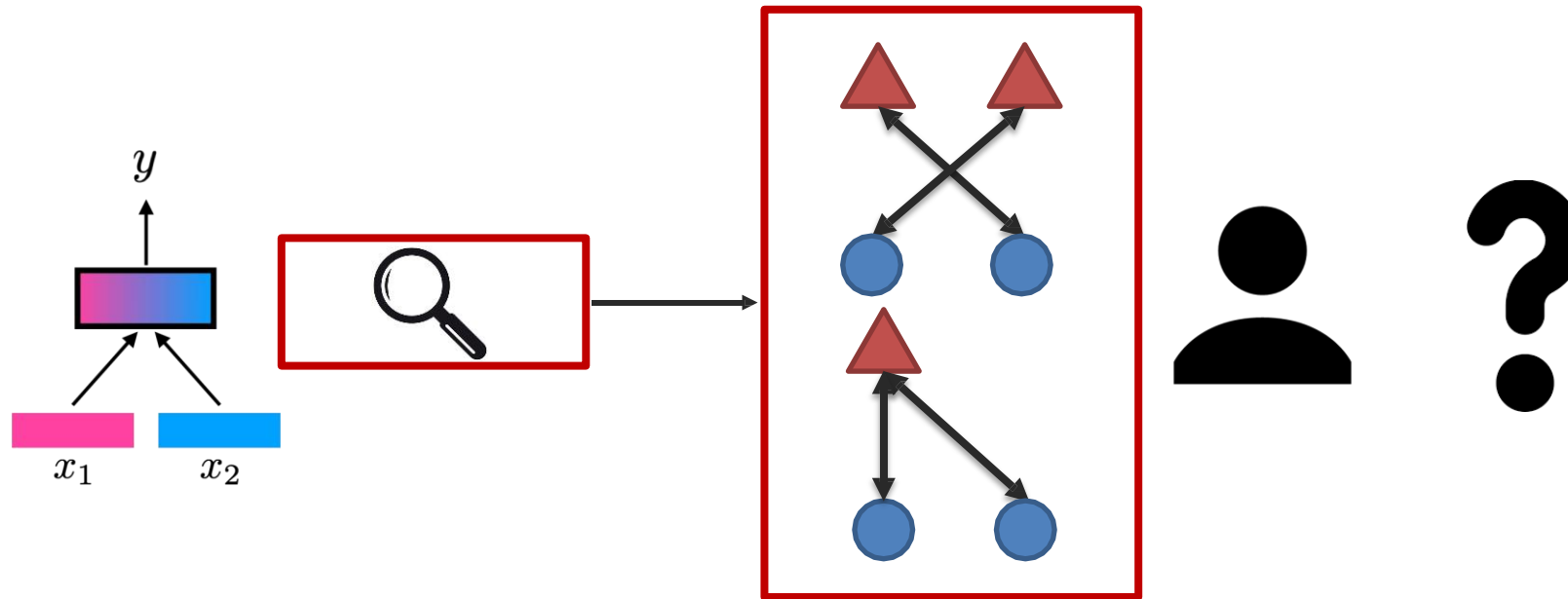
- Model selection: given fixed budget, try randomly or try models in order based on what quantification tells me.
- Data/modality selection: given fixed budget, collect random data or collect based on what quantification tells me.
- If quantification gives theoretical result, check how well the theory matches experiments.

Challenges: Quantifying Multimodal Interactions

Open
challenges

Open challenges:

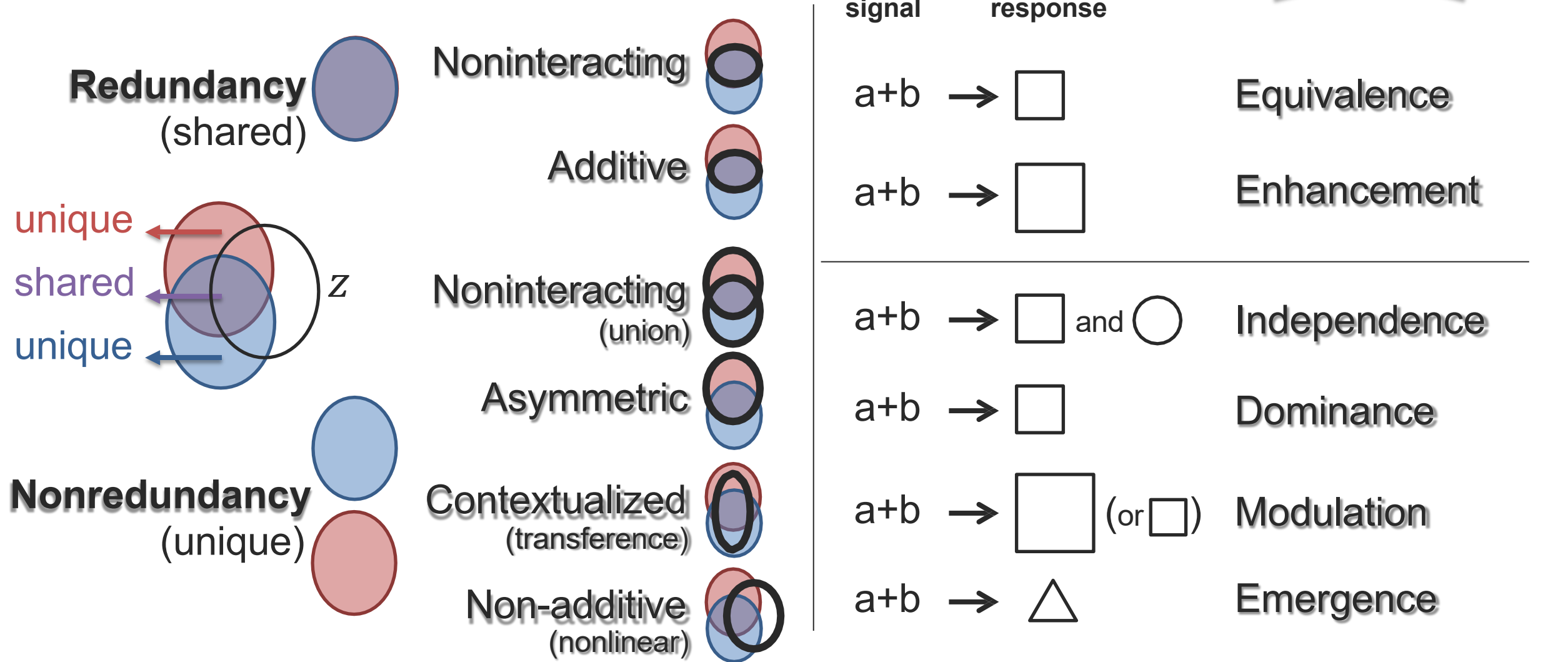
- Faithfulness: do explanations accurately reflect model's internal mechanics?
- Usefulness: unclear if explanations help humans
- Disagreement: different interpretation methods may generate different explanations
- Evaluate: how to best evaluate interpretation methods



[Chandrasekaran et al., Do explanations make VQA models more predictable to a human? EMNLP 2018]

[Krishna et al., The Disagreement Problem in Explainable Machine Learning: A Practitioner's Perspective. arXiv 2022]

Challenges: Quantifying Multimodal Interactions



Challenges: Quantifying Multimodal Interactions

Recall error analysis!

Causal, logical interactions beyond additive/multiplicative

Covariant VQA

Target object in question

Q: How many zebras are there in the picture?

A: 2

zebra removed A: 1

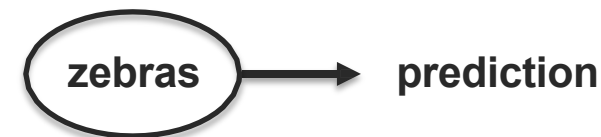


Baselines:

2

2

i.e., treatment variable



Interventional conditional: $p(y|do(zebras = 1))$

Existing models struggle to adapt to targeted causal interventions.
How can we make them more robust to spurious correlations?

Sub-Challenge 6c: Multimodal Learning Process

Definition: Characterizing the learning and optimization challenges involved when learning from heterogeneous data.

Kinetics dataset



(a) headbanging



(c) shaking hands



(e) robot dancing



(g) riding a bike



Adding more modalities should always help?

Modalities: RGB (video clips)

A (Audio features)

OF (optical flow - motion)

Dataset	Multi-modal	V@1	Best Uni	V@1	Drop
Kinetics	A + RGB	71.4	RGB	72.6	-1.2
	RGB + OF	71.3	RGB	72.6	-1.3
	A + OF	58.3	OF	62.1	-3.8
	A + RGB + OF	70.0	RGB	72.6	-2.6

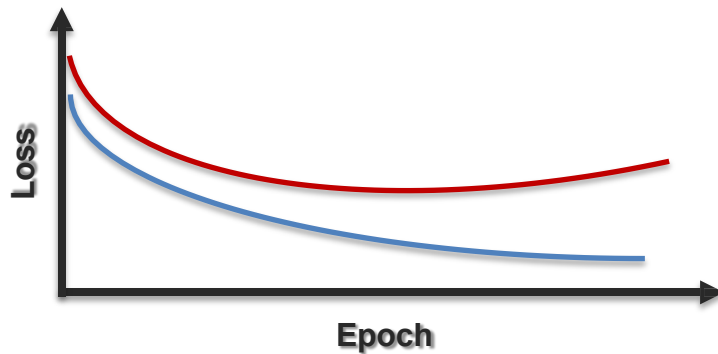
But sometimes multimodal doesn't help! **Why?**

Optimization challenges

Learning and optimization challenges

2 explanations for drop in performance:

1. Multimodal networks are more prone to overfitting due to **increased complexity**
2. Different modalities overfit and generalize at **different rates**



Key idea 1: compute overfitting-to-generalization ratio (OGR)



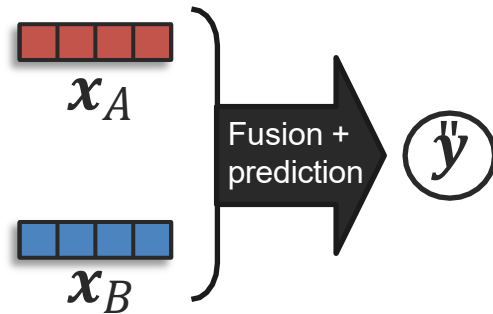
Gap between training and valid loss

OGR wrt each modality tells us
how much to train that modality

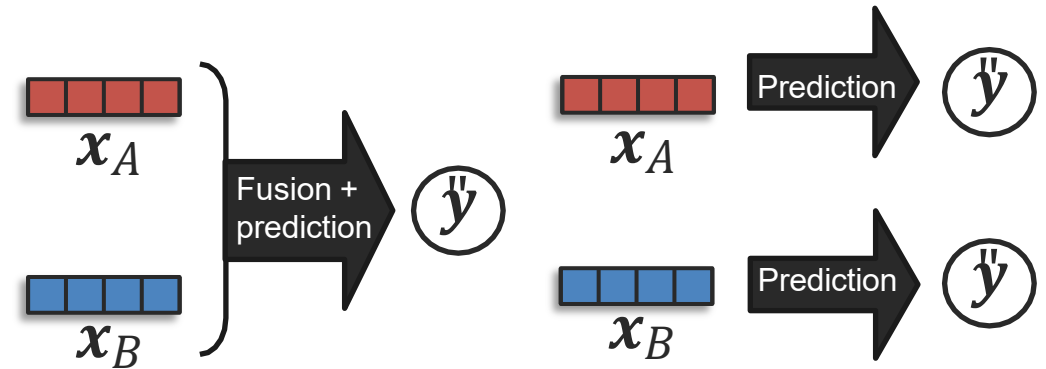
Optimization challenges

Learning and optimization challenges

Conventional approach



Proposed approach



Key idea 2: Simultaneously train unimodal networks to estimate OGR wrt each modality



Reweight multimodal loss using unimodal OGR values



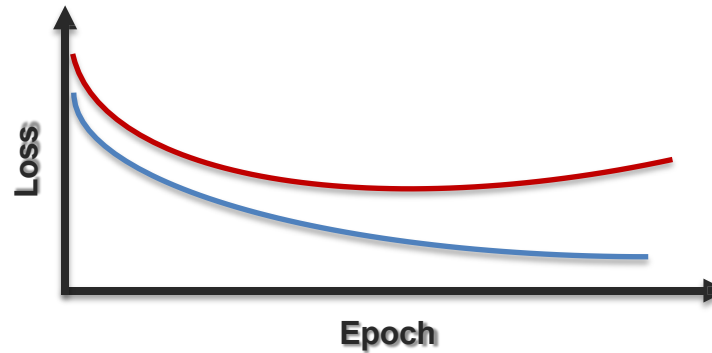
Allows to better balance generalization & overfitting rate of different modalities

Challenges

Open
challenges

Open challenges:

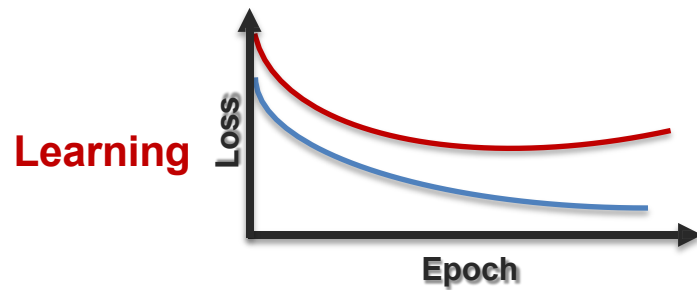
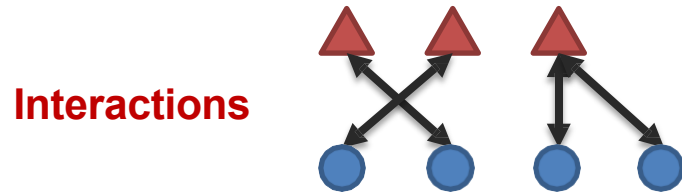
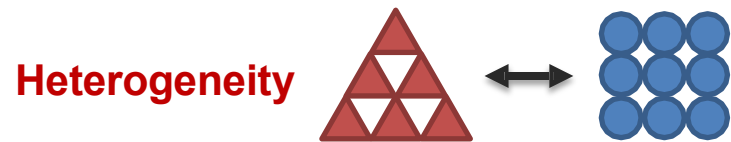
- Learning, generalization, and optimization in high-dimensional settings ($p \gg n$).
- Modality shortcuts and biases.
- Dimensionality reduction, modality selection, approximate inference.
- Reducing time and space complexity, model compression and efficiency.



More Quantification

Dimensions of quantification

Representation Alignment Reasoning Transference Generation



Conclusion

What is a Modality?

Multimodal Behaviors and Signals

Language

- **Lexicon**
 - Words
- **Syntax**
 - Part-of-speech
 - Dependencies
- **Pragmatics**
 - Discourse acts

Acoustic

- **Prosody**
 - Intonation
 - Voice quality
- **Vocal expressions**
 - Laughter, moans

Visual

- **Gestures**
 - Head gestures
 - Eye gestures
 - Arm gestures
- **Body language**
 - Body posture
 - Proxemics
- **Eye contact**
 - Head gaze
 - Eye gaze
- **Facial expressions**
 - FACS action units
 - Smile, frowning

Touch

- **Haptics**
- **Motion**

Physiological

- **Skin conductance**
- **Electrocardiogram**

Mobile

- **GPS location**
- **Accelerometer**
- **Light sensors**

What is Multimodal?

A dictionary definition...

Multimodal: with multiple modalities

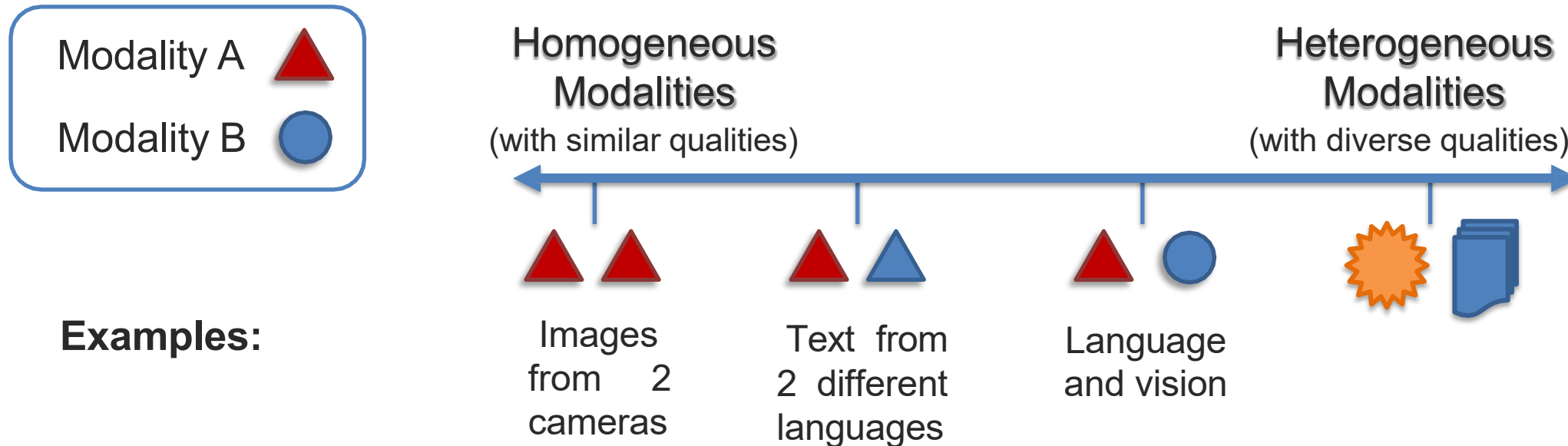
A research-oriented definition...

***Multimodal* is the scientific study of
heterogeneous and interconnected data**


Connected + Interacting

Heterogeneous Modalities

Heterogeneous: Diverse qualities, structures and representations.



Abstract modalities are more likely to be homogeneous

Connected Modalities

Connected: Shared information that relates modalities



Statistical



Association

Dependency



e.g., correlation,
co-occurrence



e.g., causal,
temporal

Semantic



Correspondence

Relationship



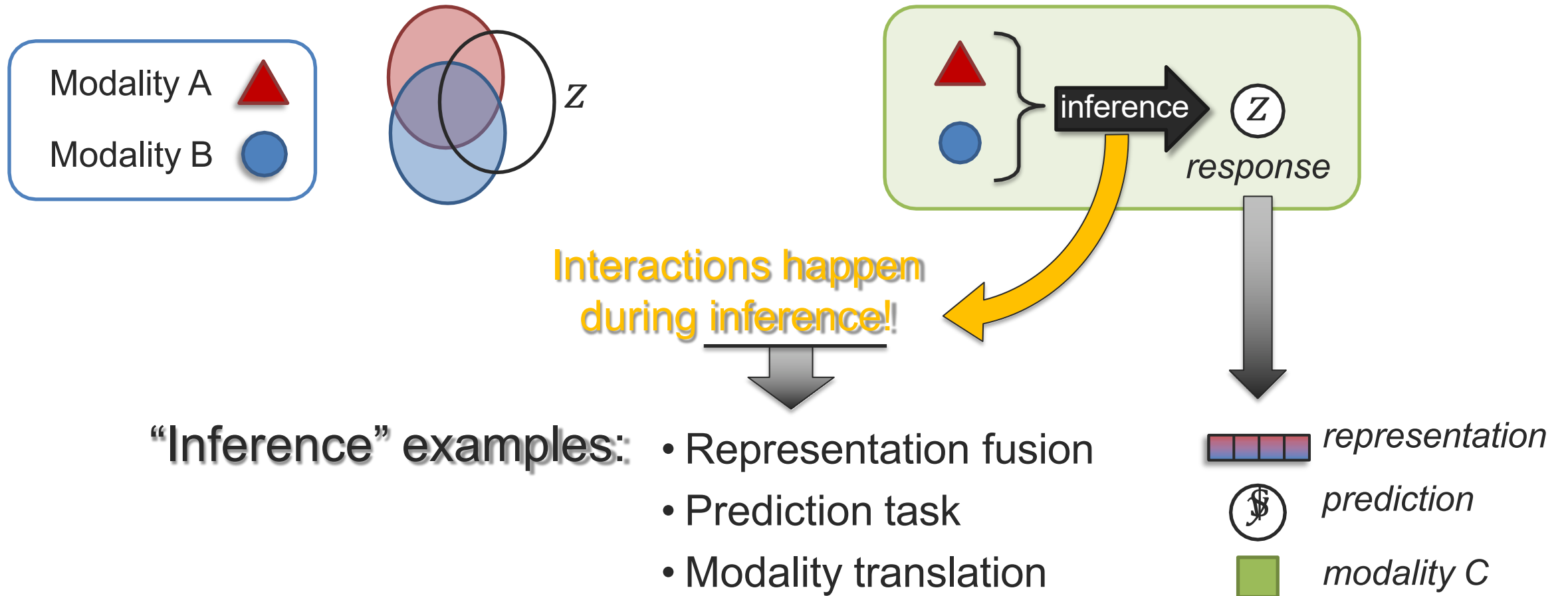
e.g., grounding



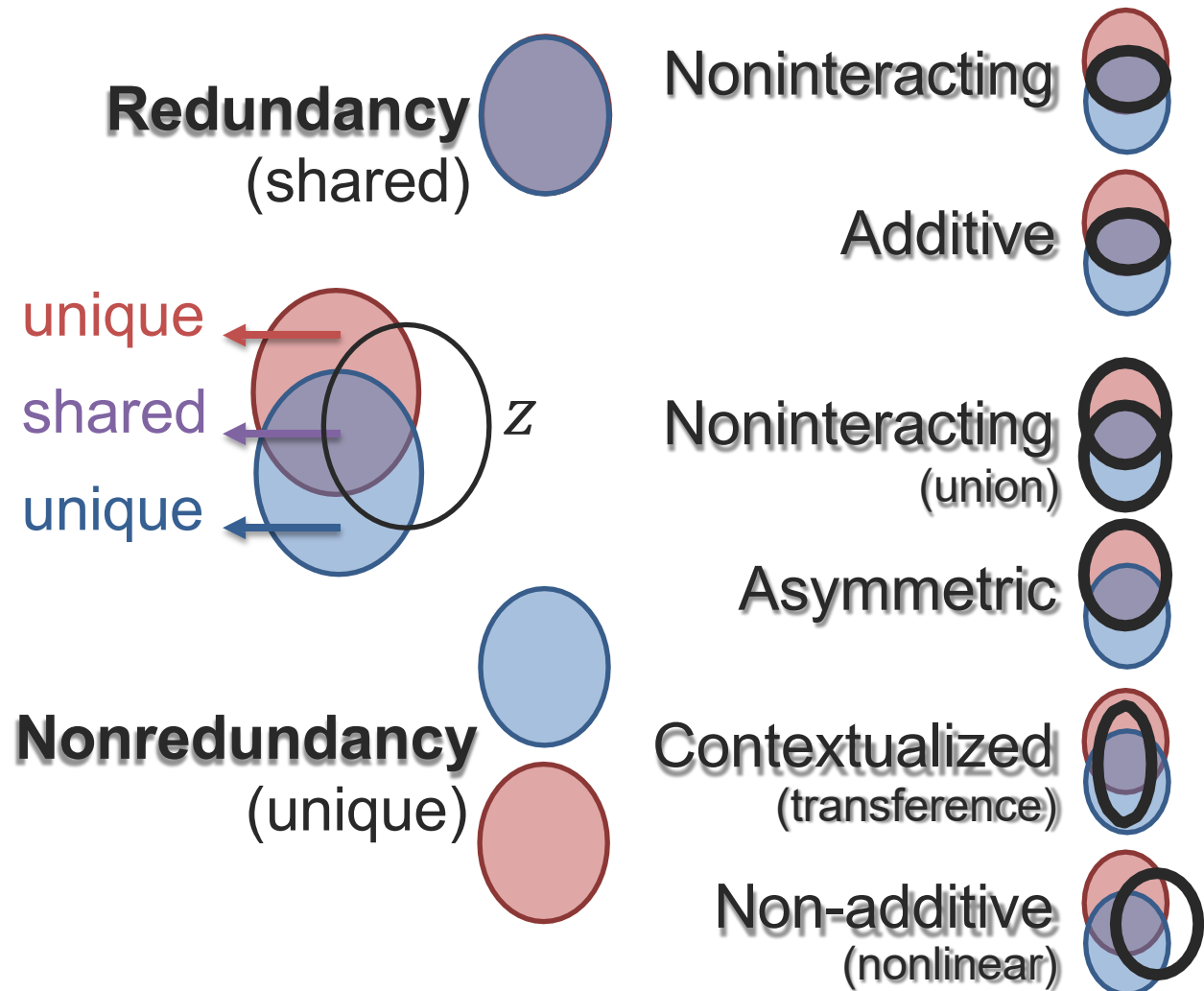
e.g., function

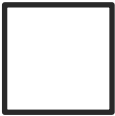
Interacting Modalities

Interacting: process affecting each modality, creating new response



Interacting Modalities



signal	response	
$a+b$	$\rightarrow$ 	Equivalence
$a+b$	$\rightarrow$ 	Enhancement
$a+b$	$\rightarrow$  and 	Independence
$a+b$	$\rightarrow$ 	Dominance
$a+b$	$\rightarrow$  (or )	Modulation
$a+b$	$\rightarrow$ 	Emergence

***What is
Multimodal?***



Why is it hard?



What is next?

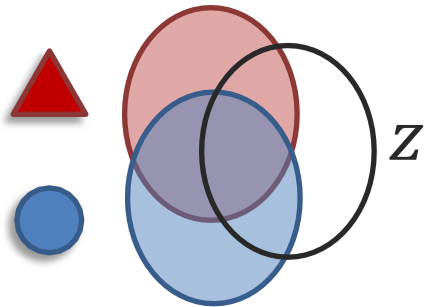
Heterogeneous



Connected

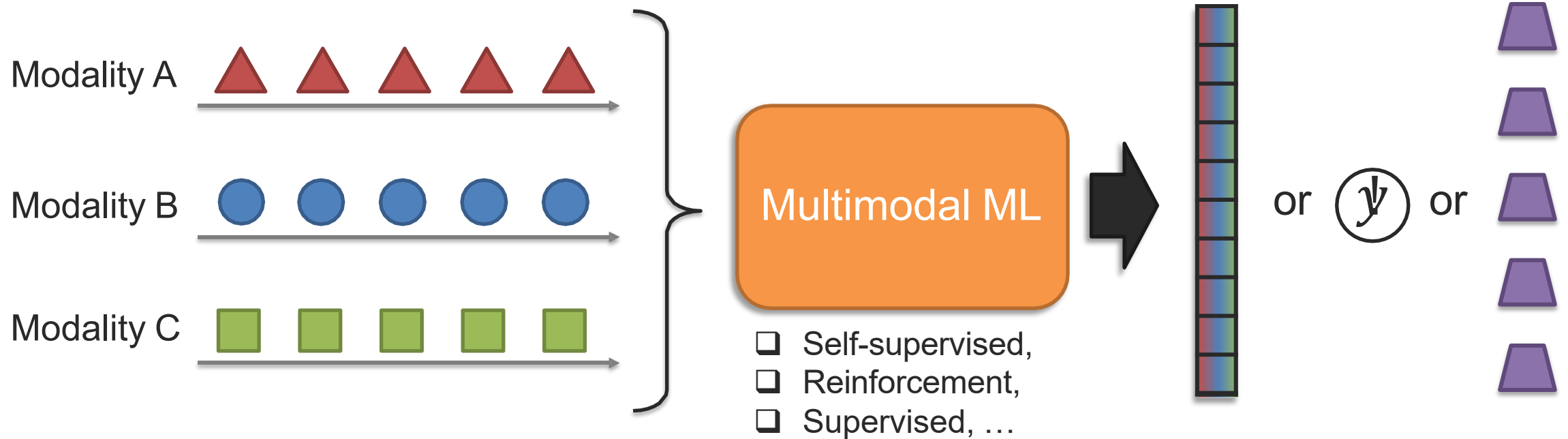


Interacting



**Multimodal is the scientific
study of heterogeneous and
interconnected data 😊**

Multimodal Machine Learning



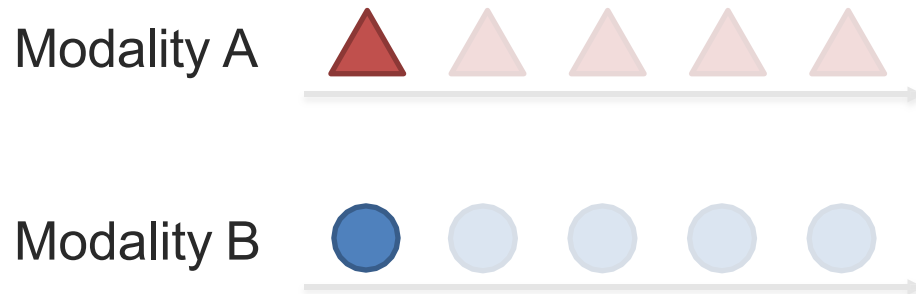
*What are the **core multimodal technical challenges**, understudied in conventional machine learning?*

Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

➡ This is a core building block for most multimodal modeling problems!

Individual elements:



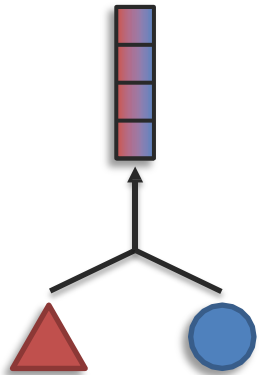
*It can be seen as a “local” representation
or
representation using holistic features*

Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

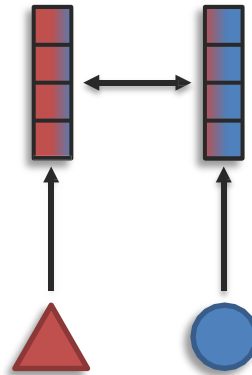
Sub-challenges:

Fusion



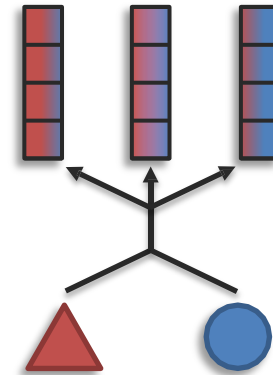
modalities $>$ # representations

Coordination



modalities $=$ # representations

Fission



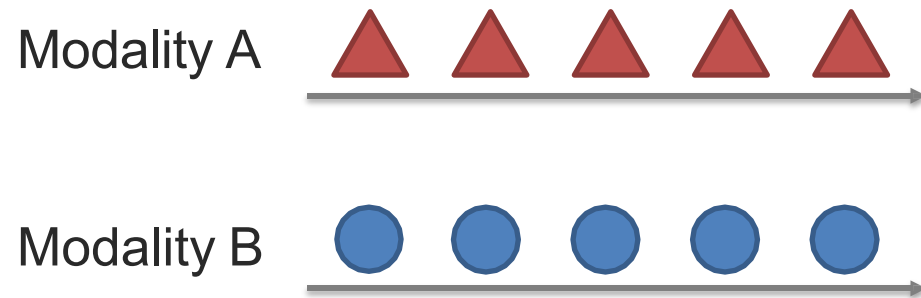
modalities $<$ # representations

Challenge 2: Alignment

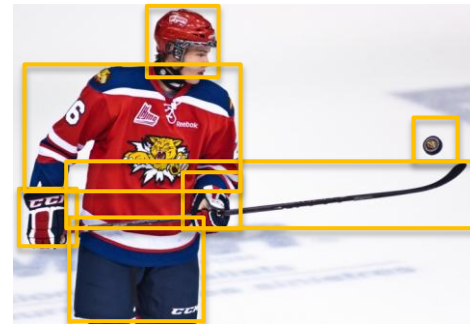
Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

➡ Most modalities have internal structure with multiple elements

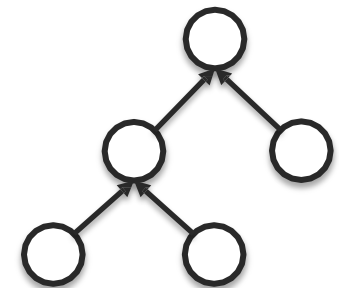
Elements with temporal structure:



Other structured examples:



Spatial



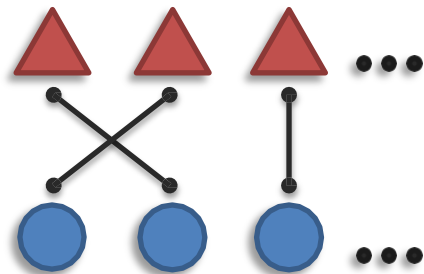
Hierarchical

Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

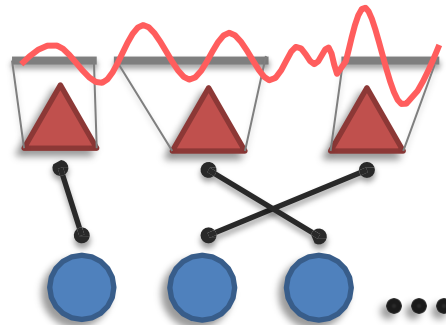
Sub-challenges:

Discrete Alignment



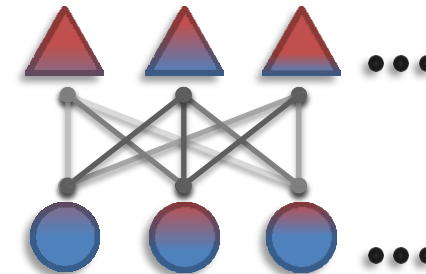
Discrete elements
and connections

Continuous Alignment



Segmentation and
continuous warping

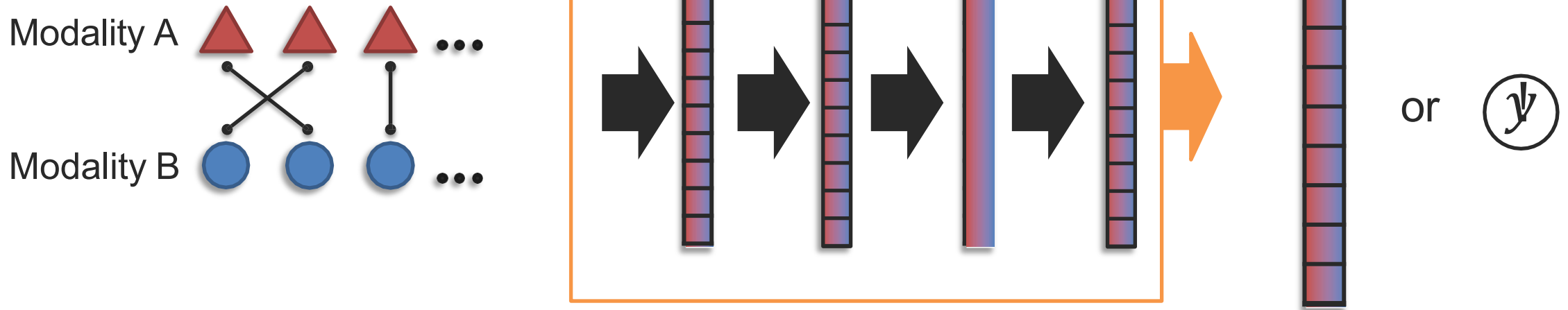
Contextualized Representation



Alignment + representation

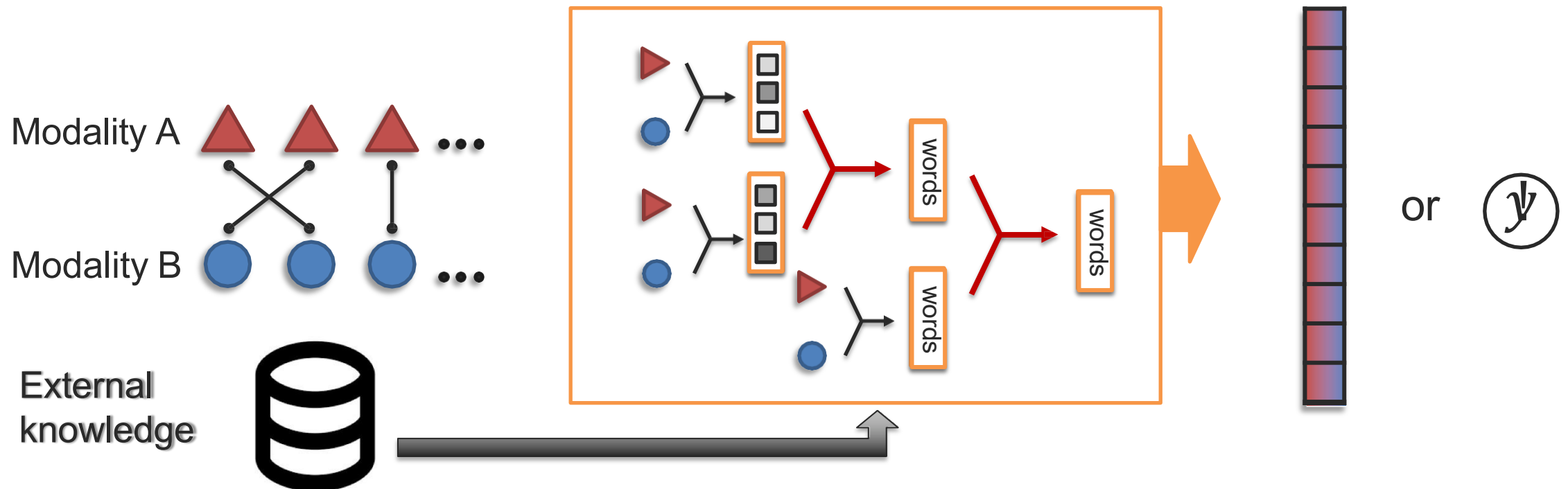
Challenge 3: Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure



Challenge 3: Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure

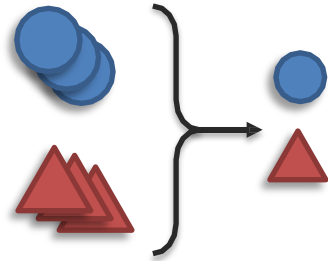


Challenge 4: Generation

Definition: Learning a generative process to produce raw modalities that reflects cross-modal interactions, structure and coherence

Sub-challenges:

Summarization



Reduction



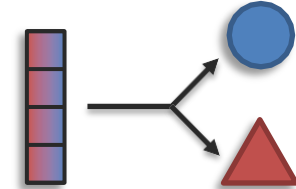
Translation



Maintenance



Creation



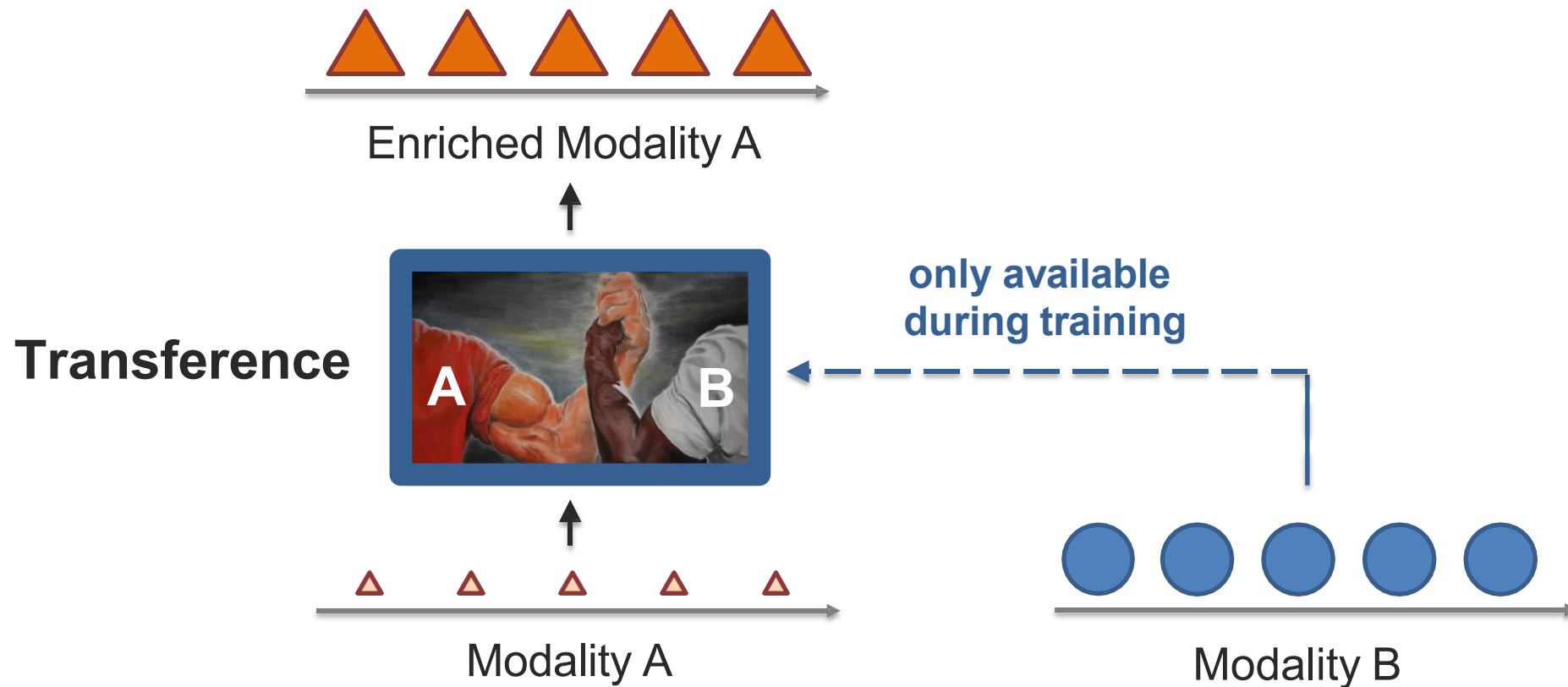
Expansion



Information:
(content)

Challenge 5: Transference

Definition: Transfer knowledge between modalities, usually to help the target modality which may be noisy or with limited resources

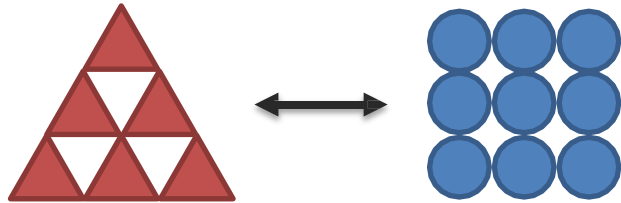


Challenge 6: Quantification

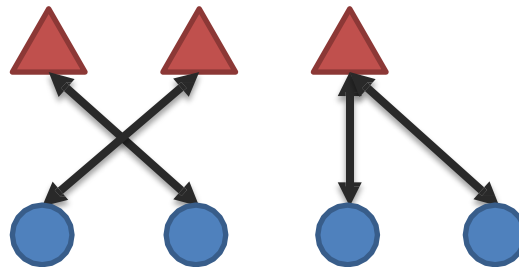
Definition: Empirical and theoretical study to better understand heterogeneity, cross-modal interactions and the multimodal learning process

Sub-challenges:

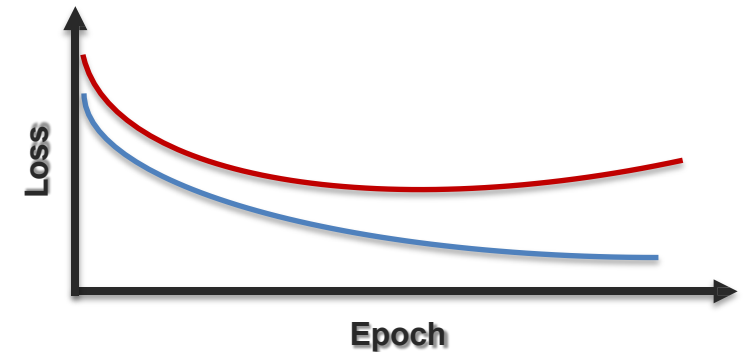
Heterogeneity



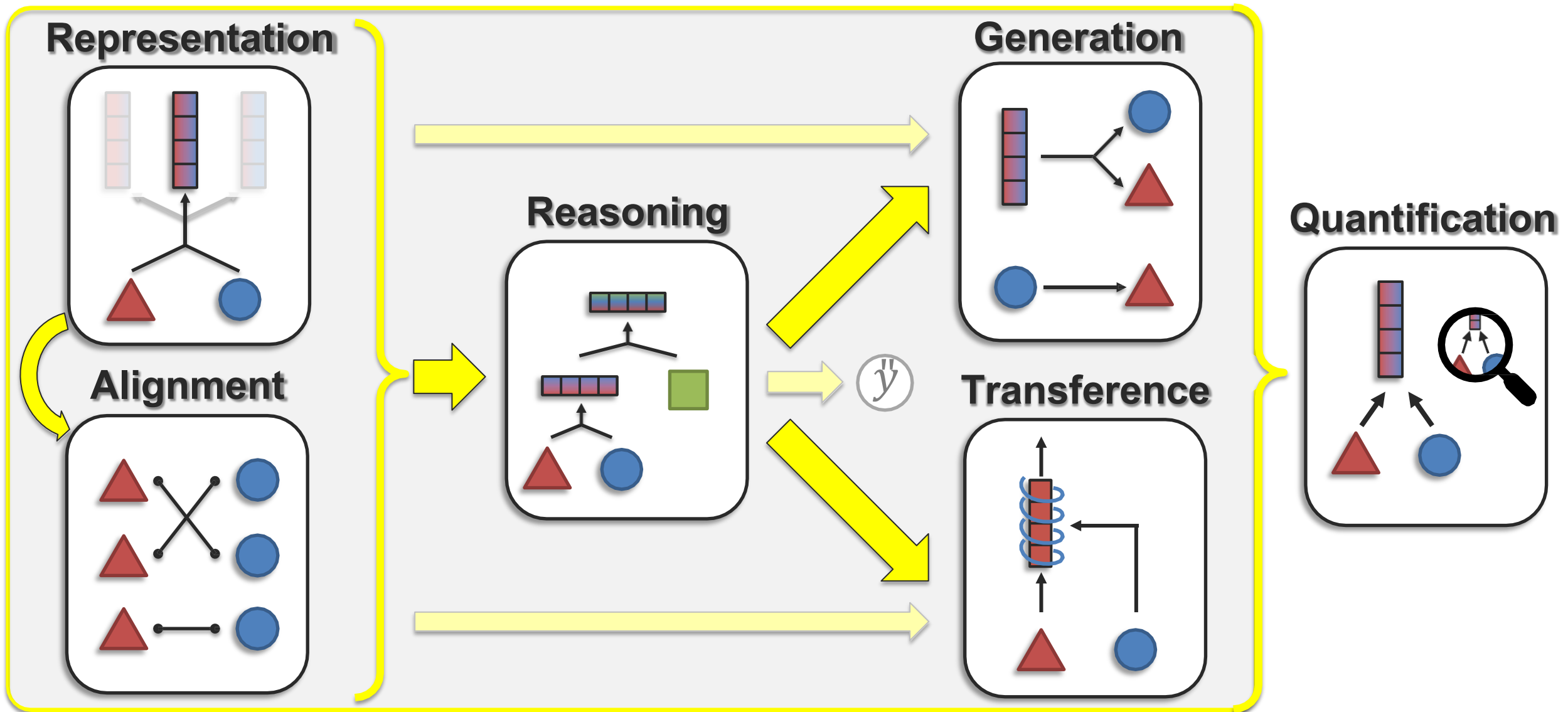
Connections & Interactions



Learning

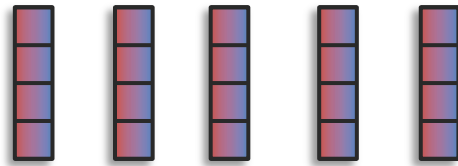


Core Multimodal Challenges



Future Direction: Heterogeneity

Homogeneity



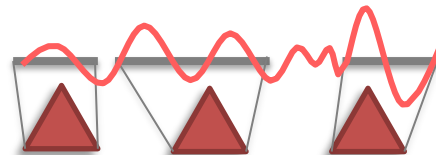
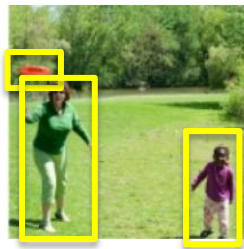
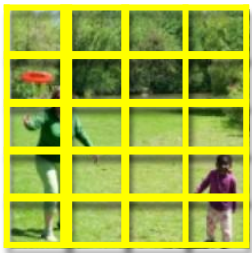
vs

Heterogeneity



Examples:

Arbitrary Tokenization



**Beyond Additive
Interactions**

Causal, logical interactions

Brain-inspired representations

MultiBench

Future Direction: High-modality

<https://github.com/pliang279/MultiBench>

Few modalities



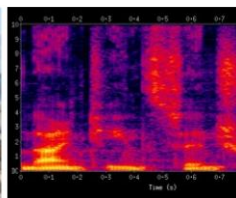
High-modality



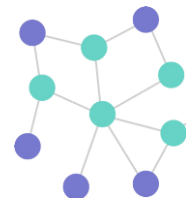
Language



Vision



Audio



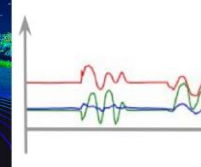
Graphs



Control



LIDAR



Sensors

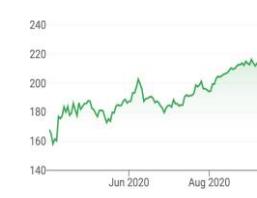


Set

SUBJECT_ID

Age
Sex
Ethnicity
...

Table



Financial



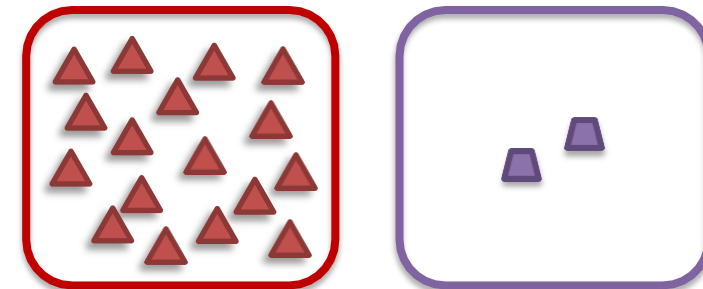
Medical

Examples:

Non-parallel learning

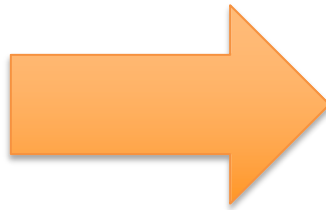
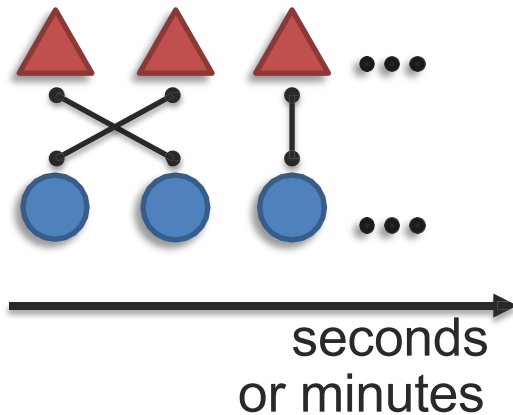


Limited resources



Future Direction: Long-term

Short-term



Long-term



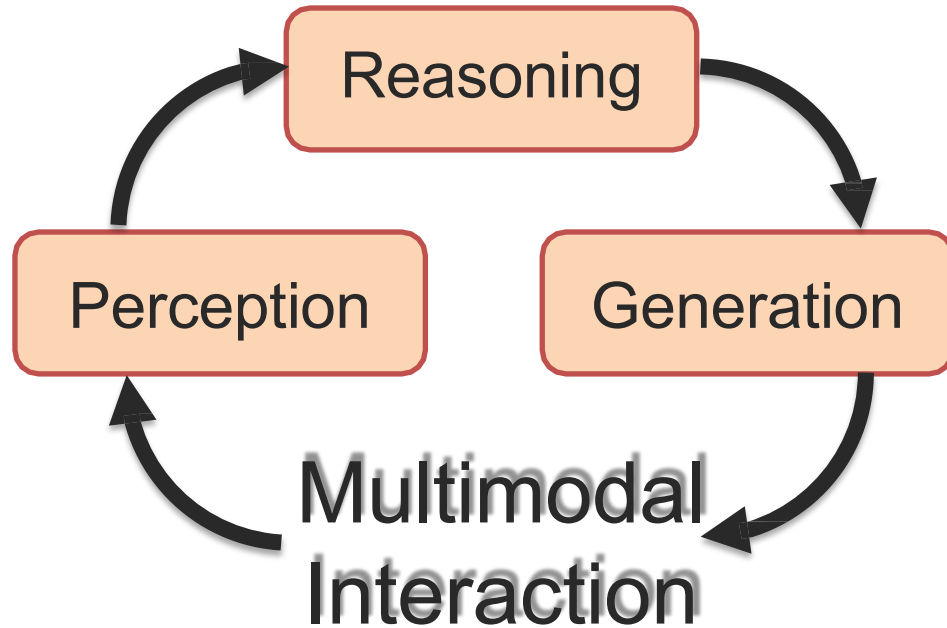
Examples:

Compositionality

Memory

Personalization

Future Direction: Interaction



Social Intelligence



Examples:

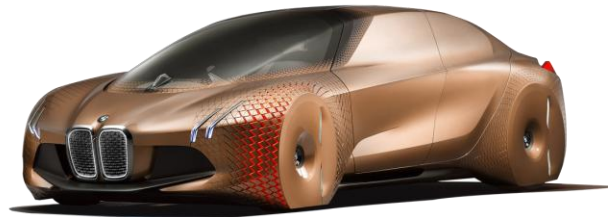
Multi-Party

Causality

Ethical



Healthcare
Decision Support



Intelligent Interfaces and
Vehicles



Online Learning
and Education

Examples:

Robustness

Fairness

Generalization

What is Multimodal?



Why is it hard?



What is next?

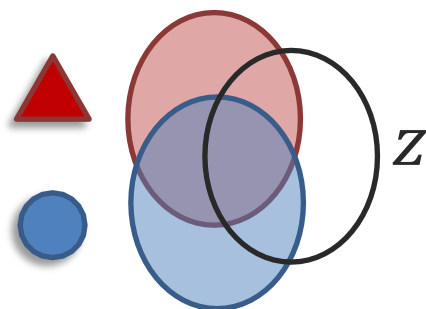
Heterogeneous



Connected



Interacting



Representation

Alignment

Reasoning

Generation

Transference

Quantification

Heterogeneity

High-modality

Long-term

Interaction

Real-world